MATHEMATICS

BASIC 2

THE LEARNER’S DISCOVERY GUIDE

29 illustrated lessons | Worked examples | Classroom projects | 435 digital practice questions

Original learner content aligned to the supplied 2019 Ghana standards-based Mathematics curriculum.

Includes companion activities for the offline Maths Quest course.

Welcome, young mathematician!

You will learn to count, compare, add, subtract, understand parts, recognise shapes, measure real objects and explain information in graphs.

How to use this book

Read the colourful lesson ideas. Study the picture and worked examples. Solve the practice questions on paper. Complete the real-life activities safely. Then try the five Play rounds and ten Challenge questions in your matching online course.

This guide follows the uploaded Basic 2 Mathematics curriculum. Where the curriculum calls for hands-on investigation, do the activity with a teacher or adult.

Learning map

NUMBER

Quest 01: Count, skip-count and estimate to 1,000

Quest 02: Explore neighbours on number charts

Quest 03: Measure and count using everyday units

Quest 04: Build numbers with hundreds, tens and ones

Quest 05: Describe numbers in equivalent ways

Quest 06: Compare and order numbers

Quest 07: Discover addition and subtraction relationships

Quest 08: Equal and not equal number sentences

Quest 09: Quick mental maths facts

Quest 10: Add and subtract using tens and ones

Quest 11: Friendly jumps and clever strategies

Quest 12: Solve everyday word problems

Quest 13: Explore halves and quarters

Quest 14: Count halves and quarters

Quest 15: Build wholes from equal parts

Quest 16: Count Ghana cedi values

ALGEBRA

Quest 17: Growing and shrinking patterns

Quest 18: Make and describe pattern rules

GEOMETRY & MEASUREMENT

Quest 19: Recognise three-dimensional objects

Quest 20: Explore flat shapes and their features

Quest 21: Build shapes from properties

Quest 22: Same length in different directions

Quest 23: Measure length, mass and capacity

Quest 24: Order objects by measurable size

Quest 25: Why standard units matter

Quest 26: Read calendars and months

Quest 27: Hours and everyday events

DATA

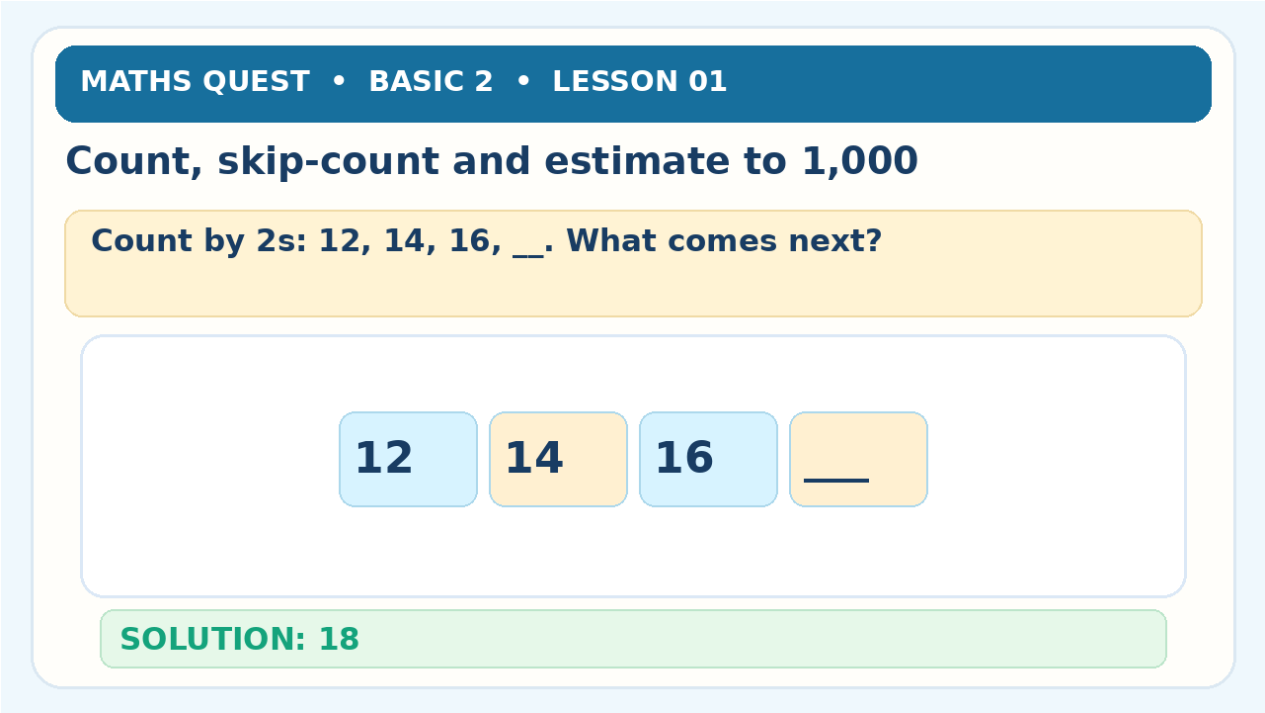
Quest 28: Collect data using tallies

Quest 29: Make and read picture graphs

Quest 01 | Count, skip-count and estimate to 1,000

YOUR MISSION

Count forwards and backwards by twos, fives and tens, and make sensible estimates up to one thousand.



LEARN IT - FOUR IMPORTANT IDEAS

1. A number tells how many objects we have. When counting a large group, place objects into smaller groups so none is missed. Ten bundles of ten make one hundred.

2. Skip-counting helps us count equal groups quickly. Counting in fives from 20 gives 20, 25, 30, 35, 40. Counting backwards in tens from 90 gives 90, 80, 70, 60.

3. An estimate is a careful guess. Look for groups of ten or one hundred before choosing an estimate. Then count or group the objects to check.

4. If a sequence misses a number, find its rule first. For example, 200, 210, 220, __, 240 uses the rule add ten, so 230 is missing.

WORKED EXAMPLES

Try this	How it works
Skip-count in fives from 25	25, 30, 35, 40, 45

Try this	How it works
What is missing: 470, 480, __, 500?	490
Estimate 9 full bundles of ten bottle tops	About 90 bottle tops

TRY IT YOURSELF

P1. Count by 2s: 12, 14, 16, __. What comes next? Answer: _____

P2. Count by 5s: 35, 40, 45, __. What is missing? Answer: _____

P3. Count by 10s: 120, 130, 140, __. What comes next? Answer: _____

P4. Count by 2s: 44, 46, 48, __. What is missing? Answer: _____

P5. Count by 5s: 75, 80, 85, __. What comes next? Answer: _____

CHALLENGE PRACTICE

C1. Count by 10s: 210, 220, 230, __. What is missing? Answer: _____

C2. Count by 2s: 56, 58, 60, __. What comes next? Answer: _____

C3. Count by 5s: 95, 100, 105, __. What is missing? Answer: _____

C4. Count by 10s: 340, 350, 360, __. What comes next? Answer: _____

C5. Count by 2s: 82, 84, 86, __. What is missing? Answer: _____

C6. Count by 5s: 155, 160, 165, __. What comes next? Answer: _____

C7. Count by 10s: 400, 410, 420, __. What is missing? Answer: _____

C8. Count by 2s: 64, 66, 68, __. What comes next? Answer: _____

C9. Count by 5s: 225, 230, 235, __. What is missing? Answer: _____

C10. Count by 10s: 600, 610, 620, __. What comes next? Answer: _____

REAL-LIFE INVESTIGATION

1. Group bottle tops in tens and count the bundles.

2. Make a classroom counting rhyme in twos and fives.

AT HOME

Ask an adult to help you estimate and count 50 dried beans.

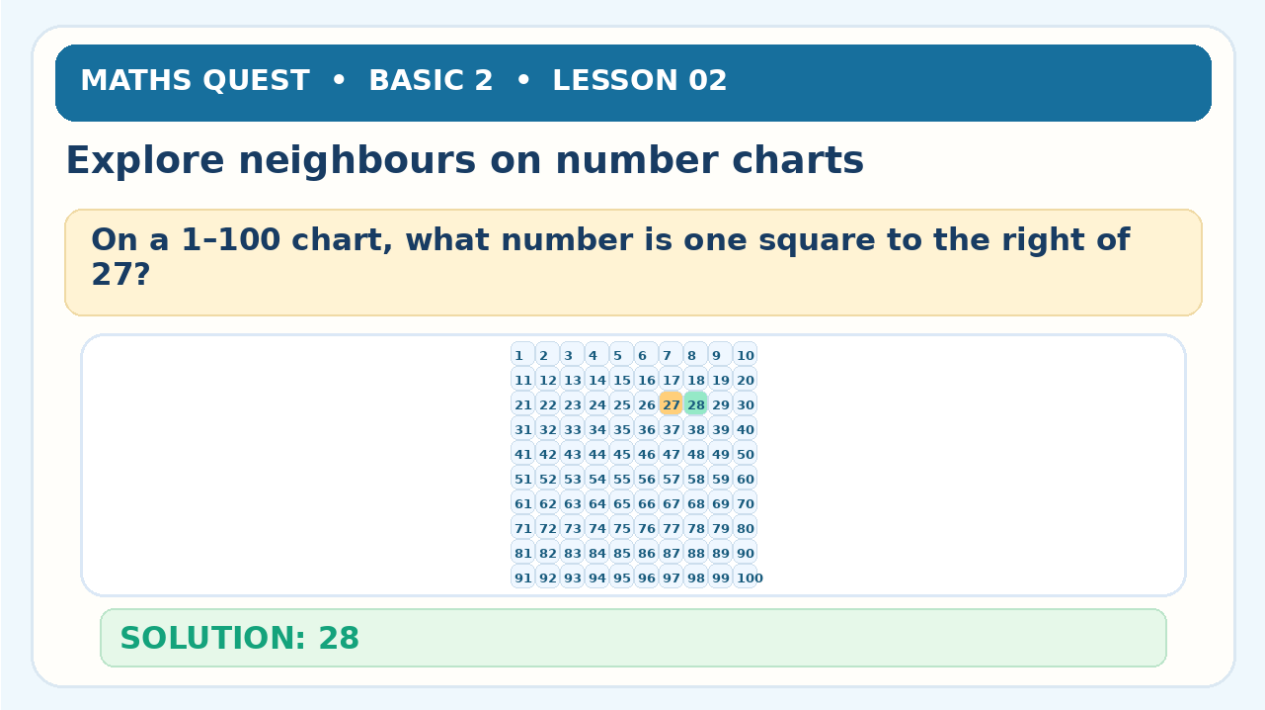
THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 02 | Explore neighbours on number charts

YOUR MISSION

Locate a number and identify numbers to its left, right, above and below in a chart.



LEARN IT - FOUR IMPORTANT IDEAS

1. A hundred chart has ten numbers across each row. Moving one square right adds one, and moving one square left subtracts one.

2. Moving one row below adds ten. Moving one row above subtracts ten. From 46, the square below is 56 and the square above is 36.

3. Some charts contain numbers above 100. Always look at how the chart is arranged before making a jump.

4. Use your finger to trace the movement and check that the numbers follow the same pattern.

WORKED EXAMPLES

Try this	How it works
Look at 27: one place right	28
Look at 27: one row below	37
Look at 52: one row above	42

TRY IT YOURSELF

P1. On a 1–100 chart, what number is one square to the right of 27? Answer: _____

P2. On a 1–100 chart, what number is one square to the left of 43? Answer: _____

P3. On a 1–100 chart, what number is one square below of 66? Answer: _____

P4. On a 1–100 chart, what number is one square above of 52? Answer: _____

P5. On a 1–100 chart, what number is one square to the right of 84? Answer: _____

CHALLENGE PRACTICE

C1. On a 1–100 chart, what number is one square to the left of 15? Answer: _____

C2. On a 1–100 chart, what number is one square below of 37? Answer: _____

C3. On a 1–100 chart, what number is one square above of 71? Answer: _____

C4. On a 1–100 chart, what number is one square to the right of 94? Answer: _____

C5. On a 1–100 chart, what number is one square to the left of 46? Answer: _____

C6. On a 1–100 chart, what number is one square below of 63? Answer: _____

C7. On a 1–100 chart, what number is one square above of 24? Answer: _____

C8. On a 1–100 chart, what number is one square to the right of 89? Answer: _____

C9. On a 1–100 chart, what number is one square to the left of 32? Answer: _____

C10. On a 1–100 chart, what number is one square below of 58? Answer: _____

REAL-LIFE INVESTIGATION

1. Draw a 10 by 10 number chart with a partner.

2. Find four neighbours around a chosen number.

AT HOME

Play chart treasure hunt with a homemade 1–100 chart.

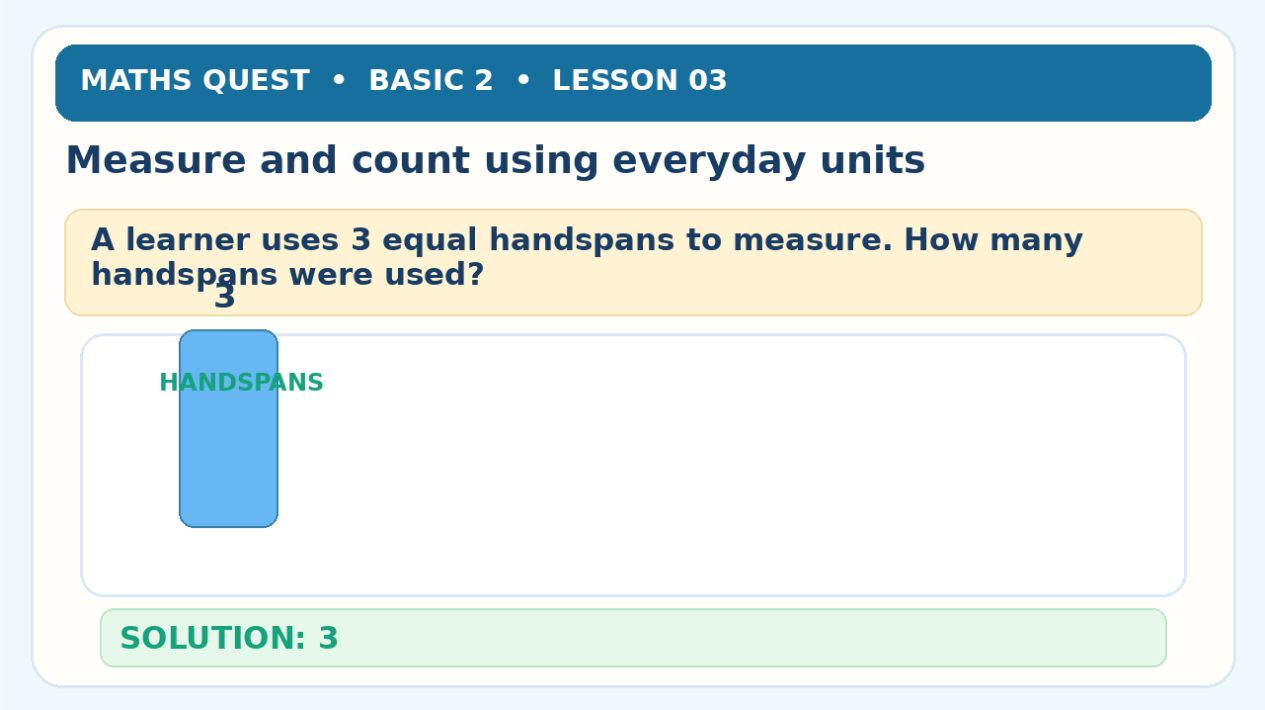
THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 03 | Measure and count using everyday units

YOUR MISSION

Use repeated handspans, strides and cupfuls to describe lengths and quantities.



LEARN IT - FOUR IMPORTANT IDEAS

1. To measure a length without a ruler, use a handspan, footstep, stick or piece of string. Repeat the same unit along the object.

2. Do not leave gaps or overlap your measuring units. If a desk is five equal handspans long, record five handspans.

3. Capacity tells us how much a container can hold. We can count equal cupfuls of water to compare containers. Use water safely with an adult.

4. Different hands and cups have different sizes, so two people may count different numbers of units for the same object.

WORKED EXAMPLES

Try this	How it works
A table is 6 equal handspans long	Record 6 handspans
A jug holds 4 equal cups of water	Record 4 cupfuls

Try this	How it works
Why use the same cup each time?	To keep the measuring unit equal

TRY IT YOURSELF

P1. A learner uses 3 equal handspans to measure. How many handspans were used? Answer: _____

P2. A learner uses 4 equal cupfuls to measure. How many cupfuls were used? Answer: _____

P3. A learner uses 5 equal paper strips to measure. How many paper strips were used? Answer: _____

P4. A learner uses 6 equal footsteps to measure. How many footsteps were used? Answer: _____

P5. A learner uses 7 equal equal blocks to measure. How many equal blocks were used? Answer: _____

CHALLENGE PRACTICE

C1. A learner uses 8 equal handspans to measure. How many handspans were used? Answer: _____

C2. A learner uses 9 equal cupfuls to measure. How many cupfuls were used? Answer: _____

C3. A learner uses 10 equal paper strips to measure. How many paper strips were used? Answer: _____

C4. A learner uses 11 equal footsteps to measure. How many footsteps were used? Answer: _____

C5. A learner uses 3 equal equal blocks to measure. How many equal blocks were used? Answer: _____

C6. A learner uses 4 equal handspans to measure. How many handspans were used? Answer: _____

C7. A learner uses 5 equal cupfuls to measure. How many cupfuls were used? Answer: _____

C8. A learner uses 6 equal paper strips to measure. How many paper strips were used? Answer: _____

C9. A learner uses 7 equal footsteps to measure. How many footsteps were used? Answer: _____

C10. A learner uses 8 equal equal blocks to measure. How many equal blocks were used? Answer: _____

REAL-LIFE INVESTIGATION

1. Measure a book using the same paper strip each time.

2. Compare the cupfuls held by two containers.

AT HOME

Measure a safe household object in handspans and record the result.

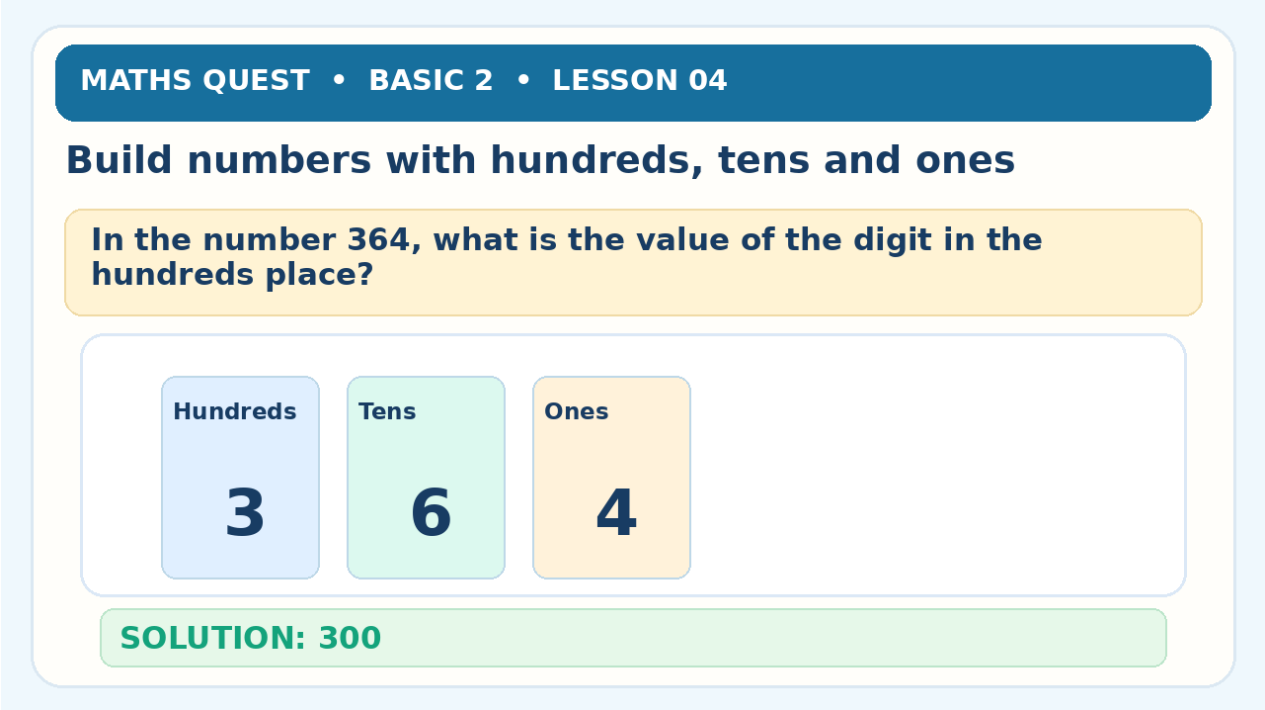
THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 04 | Build numbers with hundreds, tens and ones

YOUR MISSION

Show how the position of each digit changes its value in numbers to one thousand.



LEARN IT - FOUR IMPORTANT IDEAS

1. Every digit has a place. In 364, the 3 is in the hundreds place and means 300. The 6 means 60, and the 4 means 4.

2. A bundle of ten sticks represents one ten. Ten bundles of ten represent one hundred. Place-value blocks help us see big numbers.

3. You can break a number into parts: 364 = 300 + 60 + 4. We call this partitioning.

4. The digits 3, 6 and 4 can be rearranged to make different numbers. Changing their positions changes their values.

WORKED EXAMPLES

Try this	How it works
Expand 352	300 + 50 + 2
Value of 7 in 478	70
How many tens in 264?	6 tens in the tens place

TRY IT YOURSELF

P1. In the number 364, what is the value of the digit in the hundreds place? Answer: _____

P2. In the number 478, what is the value of the digit in the tens place? Answer: _____

P3. In the number 205, what is the value of the digit in the ones place? Answer: _____

P4. In the number 629, what is the value of the digit in the hundreds place? Answer: _____

P5. In the number 157, what is the value of the digit in the tens place? Answer: _____

CHALLENGE PRACTICE

C1. In the number 843, what is the value of the digit in the ones place? Answer: _____

C2. In the number 390, what is the value of the digit in the hundreds place? Answer: _____

C3. In the number 614, what is the value of the digit in the tens place? Answer: _____

C4. In the number 752, what is the value of the digit in the ones place? Answer: _____

C5. In the number 981, what is the value of the digit in the hundreds place? Answer: _____

C6. In the number 432, what is the value of the digit in the tens place? Answer: _____

C7. In the number 506, what is the value of the digit in the ones place? Answer: _____

C8. In the number 273, what is the value of the digit in the hundreds place? Answer: _____

C9. In the number 715, what is the value of the digit in the tens place? Answer: _____

C10. In the number 864, what is the value of the digit in the ones place? Answer: _____

REAL-LIFE INVESTIGATION

1. Make tens bundles with sticks and rubber bands.

2. Build a three-column hundreds–tens–ones mat.

AT HOME

Write five three-digit numbers and expand them.

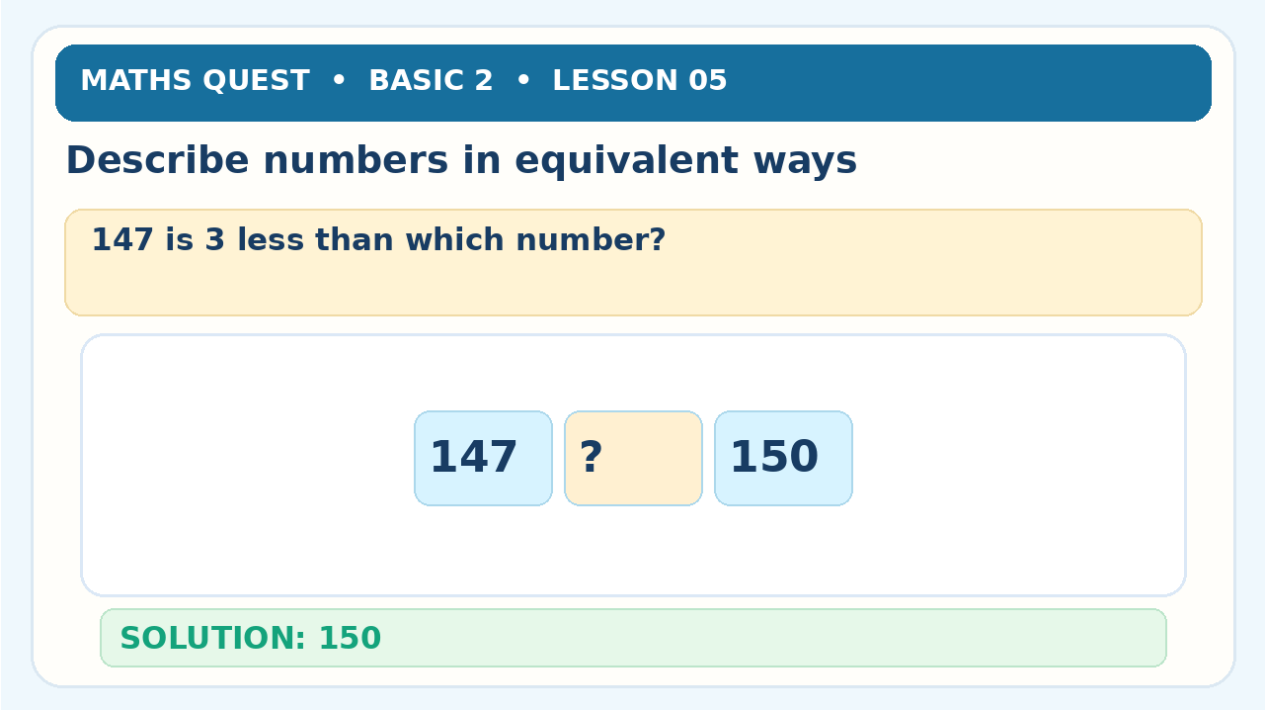
THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 05 | Describe numbers in equivalent ways

YOUR MISSION

Represent the same quantity using different but equal descriptions.



LEARN IT - FOUR IMPORTANT IDEAS

1. A number can be described in many correct ways. The quantity remains the same.

2. One hundred and forty-seven is three less than one hundred and fifty. It is also seven more than one hundred and forty.

3. A group can be rearranged into equal smaller groups. For example, twelve objects can be placed in two groups of six or three groups of four.

4. Use drawings, place-value blocks, number sentences or spoken explanations to show why two descriptions are equal.

WORKED EXAMPLES

Try this	How it works
147 is how many below 150?	3
12 objects as groups of four	3 groups
40 + 7 equals	47

TRY IT YOURSELF

P1. 147 is 3 less than which number? Answer: _____

P2. 84 is 7 more than which number? Answer: _____

P3. 235 is 10 less than which number? Answer: _____

P4. 320 is 5 more than which number? Answer: _____

P5. 96 is 2 less than which number? Answer: _____

CHALLENGE PRACTICE

C1. 183 is 3 more than which number? Answer: _____

C2. 410 is 7 less than which number? Answer: _____

C3. 68 is 10 more than which number? Answer: _____

C4. 270 is 5 less than which number? Answer: _____

C5. 152 is 2 more than which number? Answer: _____

C6. 305 is 3 less than which number? Answer: _____

C7. 73 is 7 more than which number? Answer: _____

C8. 491 is 10 less than which number? Answer: _____

C9. 186 is 5 more than which number? Answer: _____

C10. 95 is 2 less than which number? Answer: _____

REAL-LIFE INVESTIGATION

1. Arrange 12 counters in two different equal-group patterns.

2. Give a partner two descriptions of the same number.

AT HOME

Write three different true statements about 84.

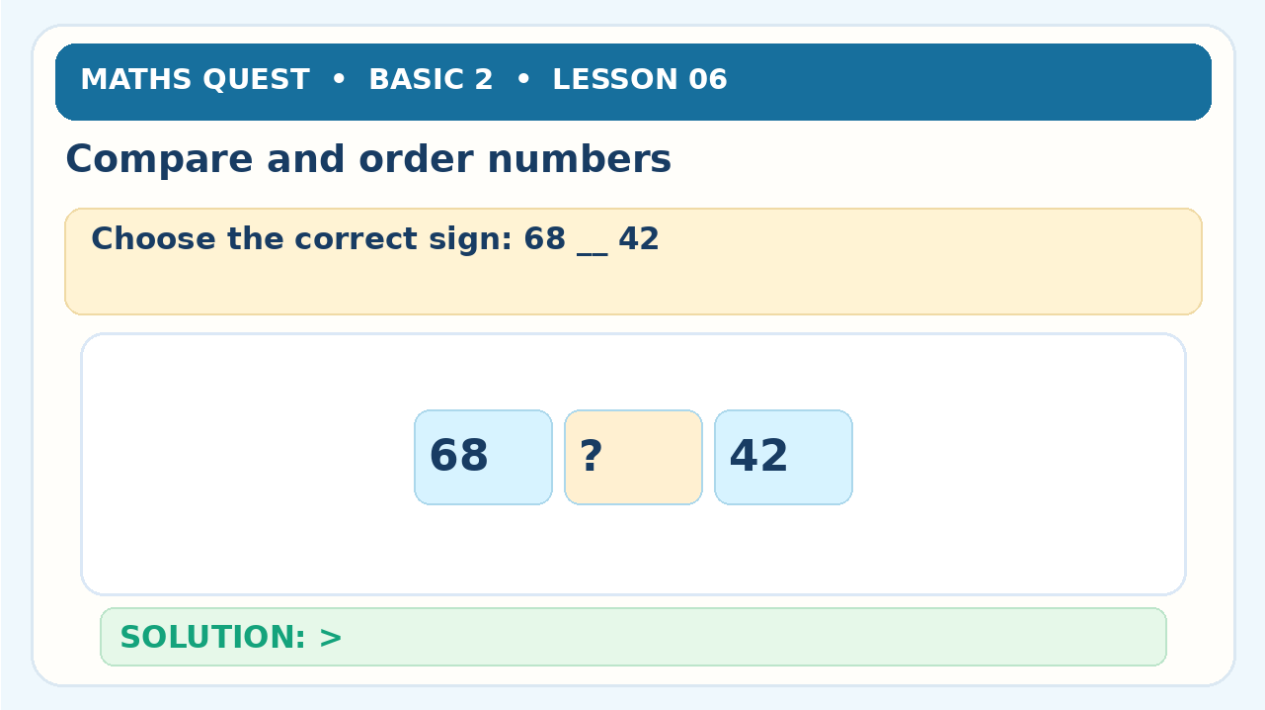
THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 06 | Compare and order numbers

YOUR MISSION

Use place value and the symbols greater than, less than and equal to when comparing numbers up to one hundred.



LEARN IT - FOUR IMPORTANT IDEAS

1. Compare tens first. For example, 68 has six tens while 42 has four tens, so 68 is greater than 42.

2. If the tens are the same, compare the ones. Since 37 and 34 both have three tens, compare seven and four.

3. The symbol > means greater than; < means less than; and = means equal to. Read from left to right.

4. To order numbers from smallest to largest, find the smallest first, then keep choosing the next larger number.

WORKED EXAMPLES

Try this	How it works
Compare 68 and 42	68 > 42
Compare 37 and 34	37 > 34

Try this	How it works
Order 18, 63, 42	18, 42, 63

TRY IT YOURSELF

P1. Choose the correct sign: 68 __ 42 Answer: _____

P2. Choose the correct sign: 37 __ 34 Answer: _____

P3. Choose the correct sign: 26 __ 58 Answer: _____

P4. Choose the correct sign: 90 __ 90 Answer: _____

P5. Choose the correct sign: 53 __ 49 Answer: _____

CHALLENGE PRACTICE

C1. Choose the correct sign: 74 __ 79 Answer: _____

C2. Choose the correct sign: 88 __ 81 Answer: _____

C3. Choose the correct sign: 62 __ 26 Answer: _____

C4. Choose the correct sign: 47 __ 47 Answer: _____

C5. Choose the correct sign: 19 __ 91 Answer: _____

C6. Choose the correct sign: 34 __ 43 Answer: _____

C7. Choose the correct sign: 55 __ 51 Answer: _____

C8. Choose the correct sign: 75 __ 85 Answer: _____

C9. Choose the correct sign: 99 __ 98 Answer: _____

C10. Choose the correct sign: 31 __ 13 Answer: _____

REAL-LIFE INVESTIGATION

1. Sort classroom number cards from least to greatest.

2. Use sticks to model a pair of two-digit numbers.

AT HOME

Order the house numbers of three imaginary houses.

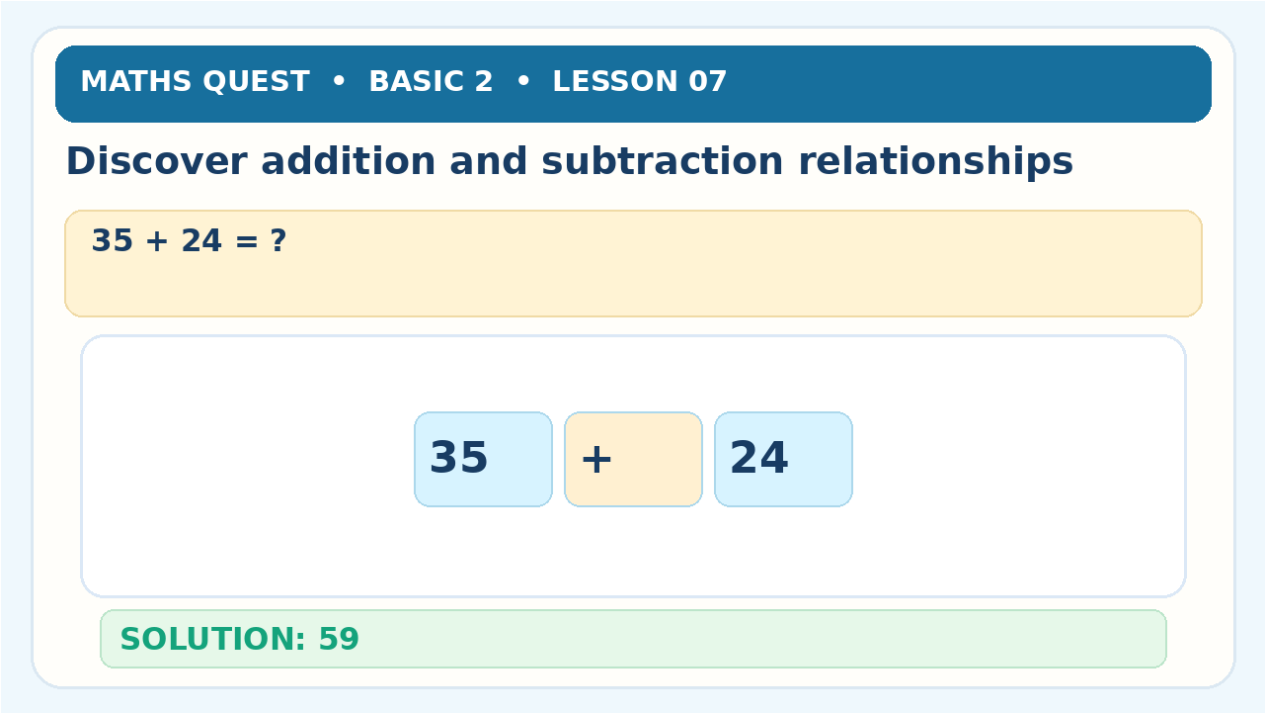
THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 07 | Discover addition and subtraction relationships

YOUR MISSION

Explain adding and subtracting to one hundred, including the effects of adding or taking away zero.



LEARN IT - FOUR IMPORTANT IDEAS

1. Addition joins quantities. We can add 35 + 24 by joining three tens and two tens, then five ones and four ones.

2. We may swap the order when adding: 12 + 7 and 7 + 12 both give 19.

3. Adding zero does not change a number. Taking away zero also leaves the number unchanged.

4. A missing number can be found with its partner operation. If 34 + ___ = 57, subtract 34 from 57 to find the missing part.

WORKED EXAMPLES

Try this	How it works
35 + 24	59
27 + 0	27
34 + ___ = 57	23

TRY IT YOURSELF

P1. 35 + 24 = ? Answer: _____

P2. 18 + 0 = ? Answer: _____

P3. 23 + 17 = ? Answer: _____

P4. 44 - 0 = ? Answer: _____

P5. 27 + 0 = ? Answer: _____

CHALLENGE PRACTICE

C1. 55 + 0 = ? Answer: _____

C2. 47 + 0 = ? Answer: _____

C3. 16 + 33 = ? Answer: _____

C4. 14 - 0 = ? Answer: _____

C5. 34 + 23 = ? Answer: _____

C6. 62 + 0 = ? Answer: _____

C7. 26 + 0 = ? Answer: _____

C8. 33 + 22 = ? Answer: _____

C9. 45 - 0 = ? Answer: _____

C10. 73 + 8 = ? Answer: _____

REAL-LIFE INVESTIGATION

1. Build and solve two addition stories using bottle tops.

2. Explain why 14 + 8 and 8 + 14 have the same answer.

AT HOME

Make a short shopping addition problem for an adult to solve.

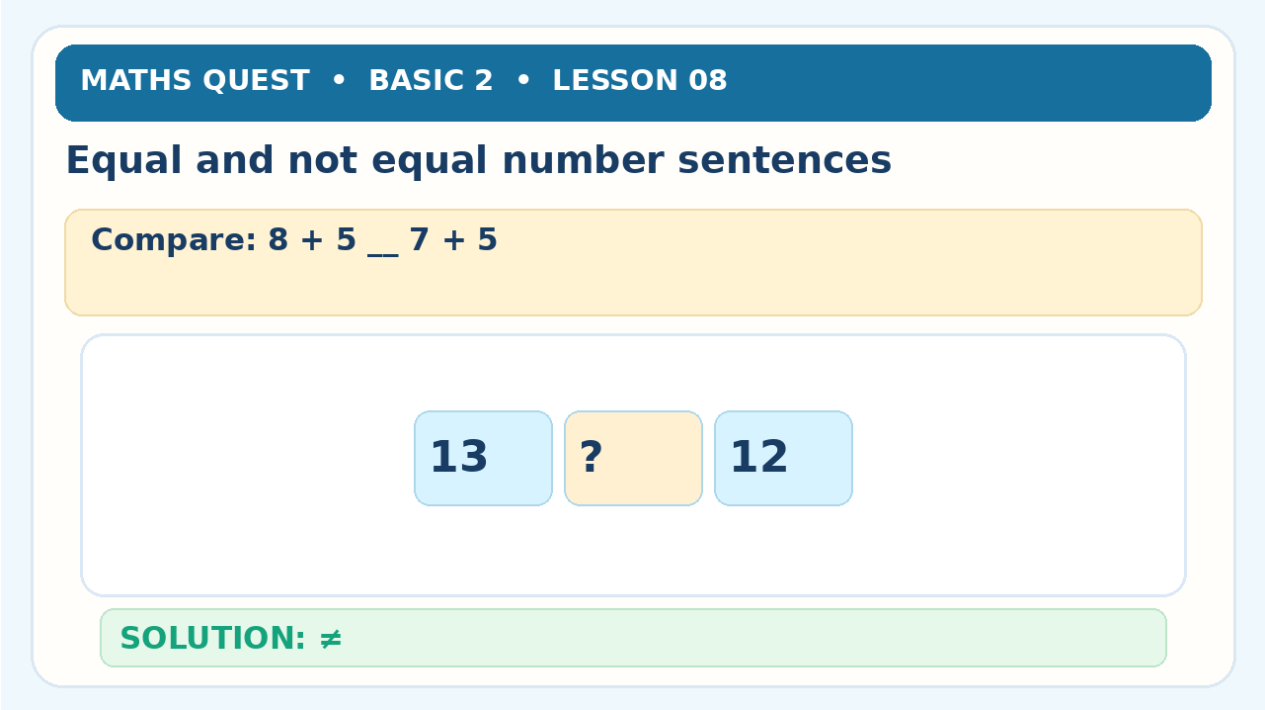
THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 08 | Equal and not equal number sentences

YOUR MISSION

Decide whether two sets or number sentences are equal and identify missing parts.



LEARN IT - FOUR IMPORTANT IDEAS

1. Equal means the two sides have the same value. The sign = shows that balance.

2. Not equal means the values are different. The sign ≠ shows they do not match.

3. Work out both sides before choosing a sign. For example, 8 + 5 is 13, but 7 + 5 is 12, so they are not equal.

4. A blank box may hide a number. Use addition or subtraction to find the number that makes both sides equal.

WORKED EXAMPLES

Try this	How it works
8 + 5 and 7 + 5	Not equal
12 + 3 and 10 + 5	Equal
9 + □ = 15	6

TRY IT YOURSELF

P1. Compare: 8 + 5 __ 7 + 5 Answer: _____

P2. Compare: 12 + 3 __ 10 + 5 Answer: _____

P3. Compare: 15 + 2 __ 9 + 8 Answer: _____

P4. Compare: 6 + 9 __ 7 + 8 Answer: _____

P5. Compare: 30 + 10 __ 20 + 20 Answer: _____

CHALLENGE PRACTICE

C1. Compare: 21 + 3 __ 18 + 5 Answer: _____

C2. Compare: 14 + 7 __ 17 + 4 Answer: _____

C3. Compare: 33 + 4 __ 30 + 7 Answer: _____

C4. Compare: 18 + 9 __ 25 + 2 Answer: _____

C5. Compare: 40 + 6 __ 40 + 5 Answer: _____

C6. Compare: 11 + 8 __ 10 + 9 Answer: _____

C7. Compare: 27 + 12 __ 29 + 10 Answer: _____

C8. Compare: 13 + 17 __ 25 + 5 Answer: _____

C9. Compare: 12 + 12 __ 15 + 9 Answer: _____

C10. Compare: 31 + 20 __ 29 + 21 Answer: _____

REAL-LIFE INVESTIGATION

1. Make two sets of counters that are equal, then change one.

2. Draw your own true and false number sentences.

AT HOME

Find two ways to make 20 with addition.

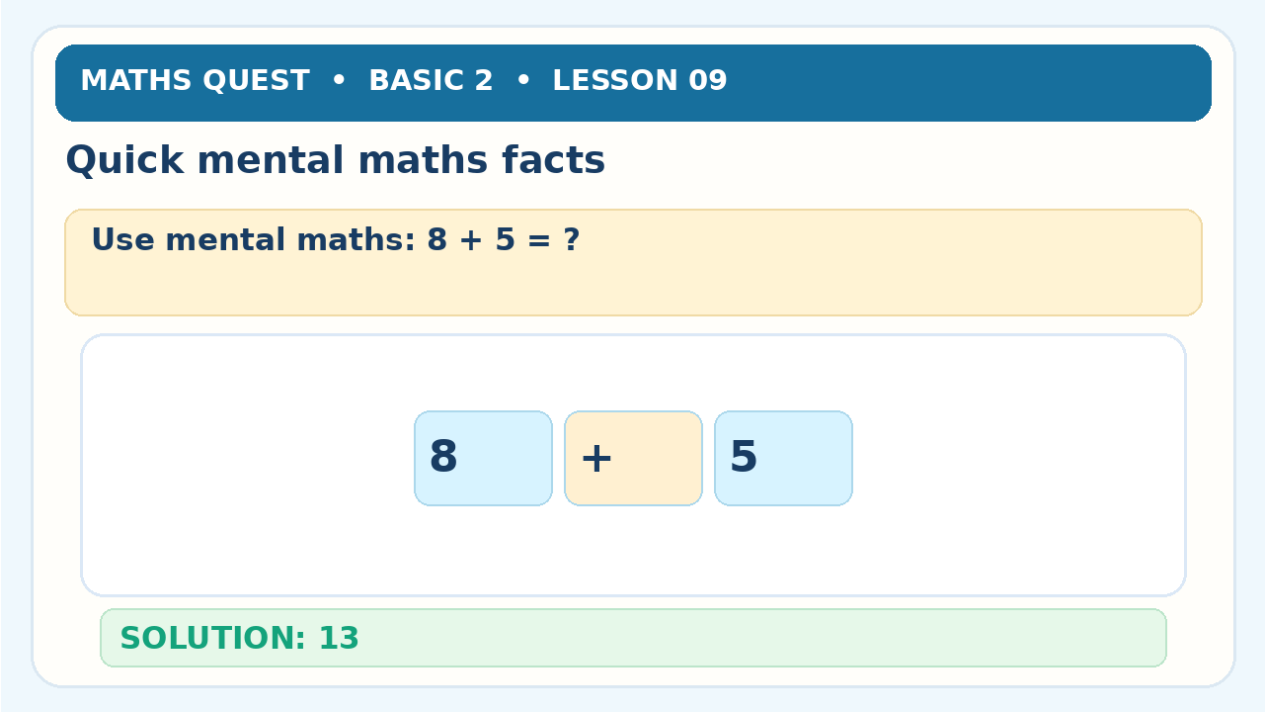
THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 09 | Quick mental maths facts

YOUR MISSION

Use doubles, making ten and related subtraction facts to solve problems mentally.



LEARN IT - FOUR IMPORTANT IDEAS

1. Mental maths means solving in your head using helpful number ideas.

2. Doubles are quick to remember. Six plus six equals twelve. For six plus seven, use six plus six plus one.

3. Make ten when you can. For 8 + 5, take 2 from 5 to make 10 + 3 = 13.

4. Subtraction can be checked using addition: if 16 - 7 = 9, then 9 + 7 must equal 16.

WORKED EXAMPLES

Try this	How it works
6 + 7 using doubles	13
8 + 5 using make ten	13
16 - 7	9

TRY IT YOURSELF

P1. Use mental maths: 8 + 5 = ? Answer: _____

P2. Use mental maths: 6 + 7 = ? Answer: _____

P3. Use mental maths: 9 + 4 = ? Answer: _____

P4. Think fast: 13 - 7 = ? Answer: _____

P5. Use mental maths: 8 + 7 = ? Answer: _____

CHALLENGE PRACTICE

C1. Use mental maths: 5 + 8 = ? Answer: _____

C2. Use mental maths: 9 + 9 = ? Answer: _____

C3. Think fast: 11 - 4 = ? Answer: _____

C4. Use mental maths: 3 + 8 = ? Answer: _____

C5. Use mental maths: 6 + 6 = ? Answer: _____

C6. Use mental maths: 7 + 7 = ? Answer: _____

C7. Think fast: 10 - 5 = ? Answer: _____

C8. Use mental maths: 9 + 8 = ? Answer: _____

C9. Use mental maths: 8 + 8 = ? Answer: _____

C10. Use mental maths: 2 + 9 = ? Answer: _____

REAL-LIFE INVESTIGATION

1. Practise quick doubles using ten fingers.

2. Play a five-minute number facts game with a partner.

AT HOME

Recall ten addition facts and check with counters.

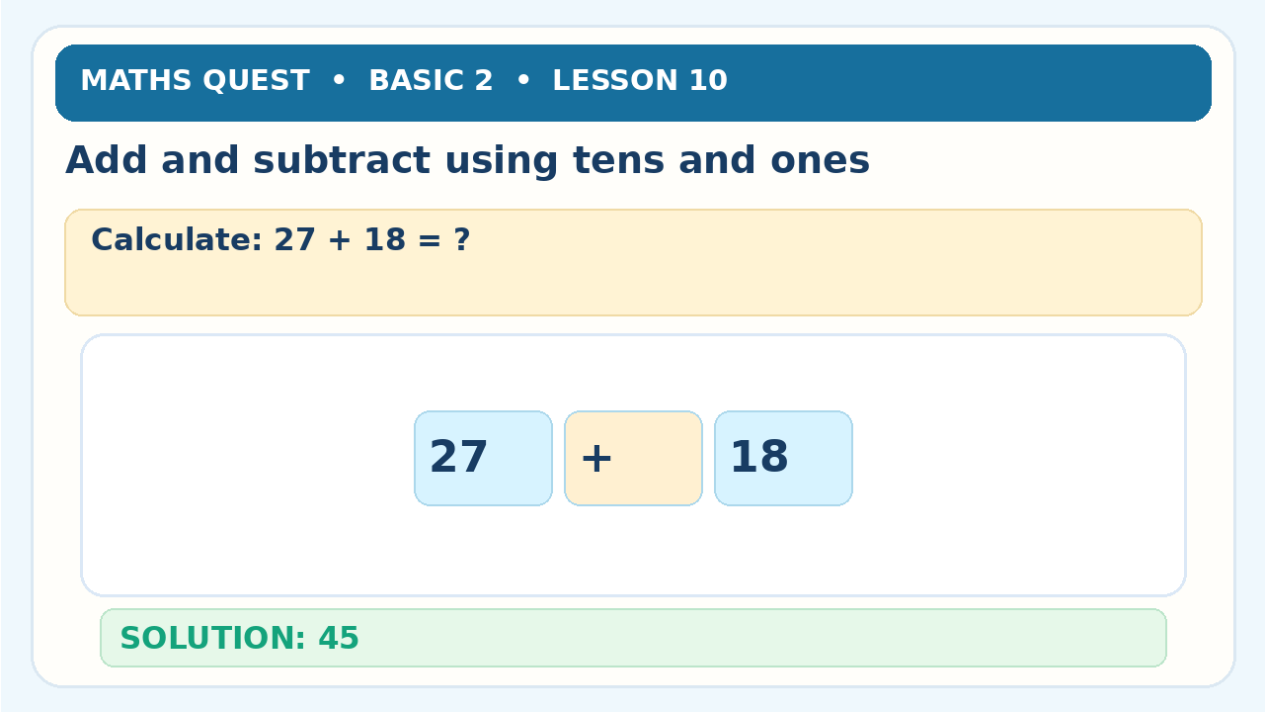
THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 10 | Add and subtract using tens and ones

YOUR MISSION

Use place-value models to add and subtract within one hundred, including regrouping.



LEARN IT - FOUR IMPORTANT IDEAS

1. Start with the ones column. Then work with the tens. Keep digits lined up in their correct places.

2. When ten or more ones appear, regroup ten ones as one ten. For 27 + 18, five ones remain and one ten joins the tens.

3. When subtracting and you do not have enough ones, exchange one ten for ten ones.

4. Check your calculation using the opposite operation or a quick estimate.

WORKED EXAMPLES

Try this	How it works
27 + 18	45
52 - 26	26
36 + 27	63

TRY IT YOURSELF

P1. Calculate: 27 + 18 = ? Answer: _____

P2. Calculate: 52 - 26 = ? Answer: _____

P3. Calculate: 36 + 27 = ? Answer: _____

P4. Calculate: 43 + 28 = ? Answer: _____

P5. Calculate: 64 - 37 = ? Answer: _____

CHALLENGE PRACTICE

C1. Calculate: 58 + 17 = ? Answer: _____

C2. Calculate: 82 - 45 = ? Answer: _____

C3. Calculate: 34 + 39 = ? Answer: _____

C4. Calculate: 75 - 48 = ? Answer: _____

C5. Calculate: 46 + 27 = ? Answer: _____

C6. Calculate: 61 - 24 = ? Answer: _____

C7. Calculate: 19 + 56 = ? Answer: _____

C8. Calculate: 90 - 37 = ? Answer: _____

C9. Calculate: 28 + 43 = ? Answer: _____

C10. Calculate: 57 - 29 = ? Answer: _____

REAL-LIFE INVESTIGATION

1. Use sticks to show an exchange of ten ones for one ten.

2. Draw addition and subtraction frames with columns.

AT HOME

Solve 47 + 26 and 83 - 35 on paper.

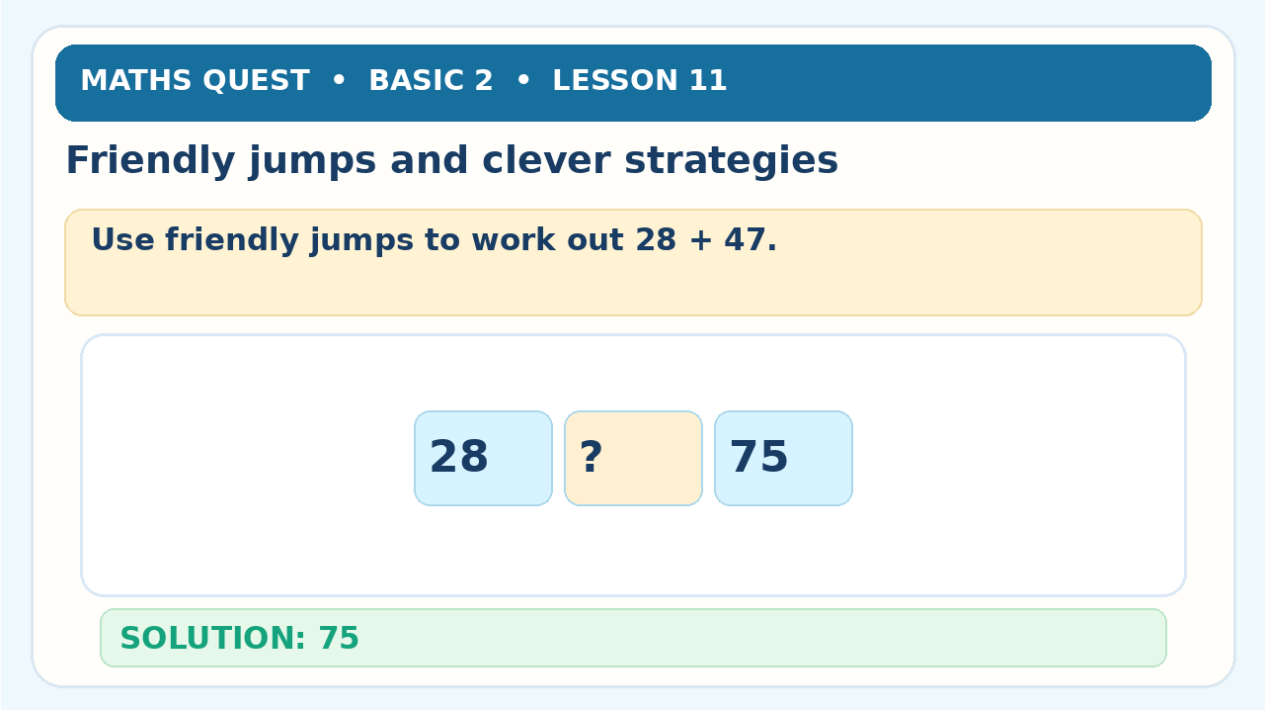
THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 11 | Friendly jumps and clever strategies

YOUR MISSION

Break apart numbers or use friendly jumps to add and subtract within one hundred.



LEARN IT - FOUR IMPORTANT IDEAS

1. You can solve a problem in more than one way. Choose a strategy you understand.

2. Break apart 28 + 47: add 20 + 40 = 60, then 8 + 7 = 15; altogether 75.

3. A friendly jump adds a ten first. For 26 + 32, add ten three times to reach 56, then two more to reach 58.

4. Sometimes you can move one from one number to another: 29 + 56 becomes 30 + 55, which is 85.

WORKED EXAMPLES

Try this	How it works
28 + 47	75
26 + 32	58
29 + 56	85

TRY IT YOURSELF

P1. Use friendly jumps to work out 28 + 47. Answer: _____

P2. Use friendly jumps to work out 26 + 32. Answer: _____

P3. Use friendly jumps to work out 29 + 56. Answer: _____

P4. Use friendly jumps to work out 38 + 25. Answer: _____

P5. Use friendly jumps to work out 46 + 29. Answer: _____

CHALLENGE PRACTICE

C1. Use friendly jumps to work out 17 + 63. Answer: _____

C2. Use friendly jumps to work out 24 + 35. Answer: _____

C3. Use friendly jumps to work out 39 + 42. Answer: _____

C4. Use friendly jumps to work out 55 + 27. Answer: _____

C5. Use friendly jumps to work out 61 + 18. Answer: _____

C6. Use friendly jumps to work out 37 + 24. Answer: _____

C7. Use friendly jumps to work out 49 + 31. Answer: _____

C8. Use friendly jumps to work out 57 + 23. Answer: _____

C9. Use friendly jumps to work out 36 + 44. Answer: _____

C10. Use friendly jumps to work out 58 + 19. Answer: _____

REAL-LIFE INVESTIGATION

1. Show 38 + 25 with number-line jumps.

2. Compare two different ways to solve 46 + 29.

AT HOME

Write a friendly-jump explanation for 37 + 24.

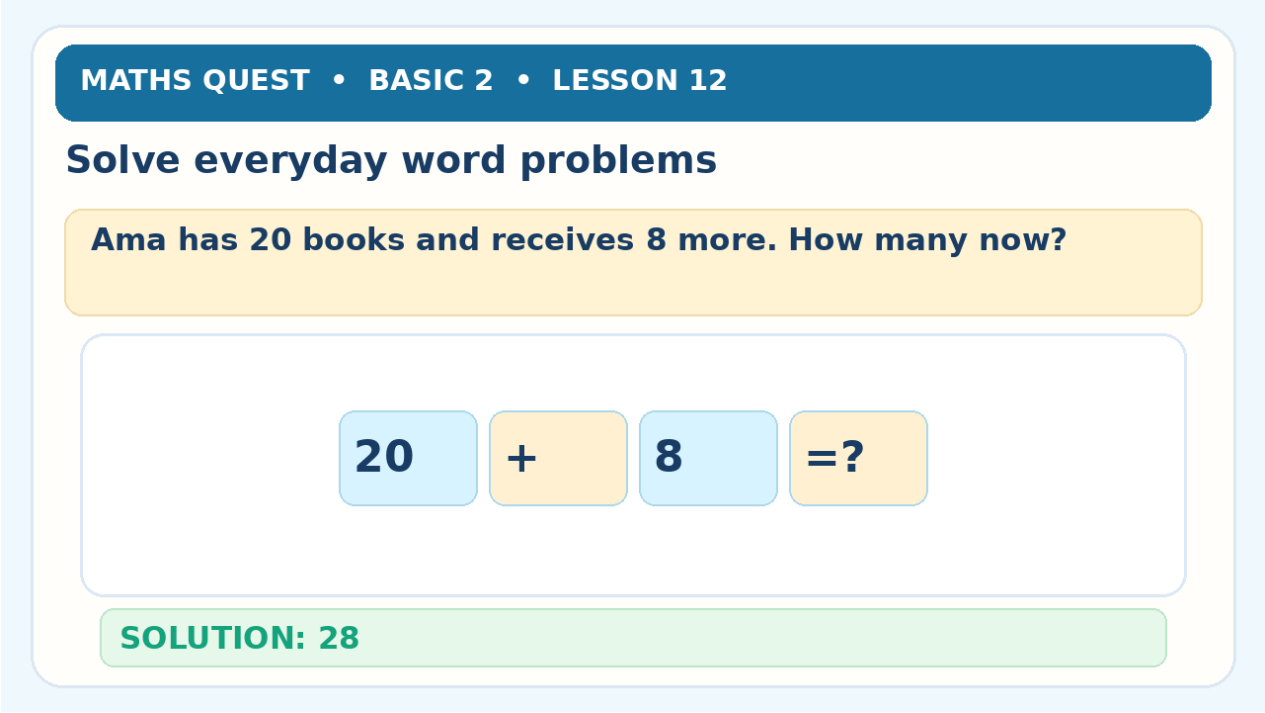
THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 12 | Solve everyday word problems

YOUR MISSION

Choose addition or subtraction to solve one-step and multi-step problems within one hundred.



LEARN IT - FOUR IMPORTANT IDEAS

1. Read the whole story first. Ask: what do I know and what must I find?

2. Words such as altogether, left, more and how many can provide clues, but always check the meaning of the story.

3. Draw a quick picture or bar model. Write the number sentence before solving.

4. For a problem with two steps, solve one part, then use that answer in the next part. Always state the answer with its unit.

WORKED EXAMPLES

Try this	How it works
Yaw has 32 books and receives 13 more	45 books
45 books; give away 11	34 books
Need GH¢8; have GH¢5	GH¢3 more

TRY IT YOURSELF

P1. Ama has 20 books and receives 8 more. How many now? Answer: _____

P2. Kofi has 31 pencils and gives away 9. How many remain? Answer: _____

P3. Abena has 24 oranges. She gets 10, then gives away 3. How many remain? Answer: _____

P4. Ama has 26 beads and receives 11 more. How many now? Answer: _____

P5. Kofi has 40 chalk sticks and gives away 12. How many remain? Answer: _____

CHALLENGE PRACTICE

C1. Abena has 30 books. She gets 13, then gives away 3. How many remain? Answer: _____

C2. Ama has 32 pencils and receives 14 more. How many now? Answer: _____

C3. Kofi has 49 oranges and gives away 15. How many remain? Answer: _____

C4. Abena has 36 beads. She gets 16, then gives away 3. How many remain? Answer: _____

C5. Ama has 38 chalk sticks and receives 8 more. How many now? Answer: _____

C6. Kofi has 49 books and gives away 9. How many remain? Answer: _____

C7. Abena has 42 pencils. She gets 10, then gives away 3. How many remain? Answer: _____

C8. Ama has 44 oranges and receives 11 more. How many now? Answer: _____

C9. Kofi has 58 beads and gives away 12. How many remain? Answer: _____

C10. Abena has 48 chalk sticks. She gets 13, then gives away 3. How many remain? Answer: _____

REAL-LIFE INVESTIGATION

1. Create a classroom books word problem.

2. Solve a two-step story with a partner and explain it.

AT HOME

Ask an adult to create a short shopping problem.

THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 13 | Explore halves and quarters

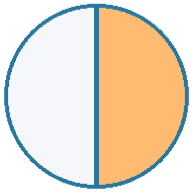
YOUR MISSION

Recognise one-half and one-quarter as one of two or four equal parts of a whole.

MATHS QUEST • BASIC 2 • LESSON 13

Explore halves and quarters

A whole is shared into two equal pieces. What is one piece called?



1 of 2 equal parts shaded

SOLUTION: one-half

LEARN IT - FOUR IMPORTANT IDEAS

1. A whole object can be split into equal parts. Equal means the parts have the same size.

2. When a whole is split into two equal parts, each part is called one-half.

3. When a whole is split into four equal parts, each part is called one-quarter or one-fourth.

4. A part is not automatically a half or a quarter if the pieces are not equal. Check the sizes before naming them.

WORKED EXAMPLES

Try this	How it works
Two equal parts of a whole	Two halves
Four equal parts of a whole	Four quarters
One of four equal pieces	One-quarter

TRY IT YOURSELF

P1. A whole is shared into two equal pieces. What is one piece called? Answer: _____

P2. A whole is shared into four equal pieces. What is one piece called? Answer: _____

P3. Are unequal pieces called halves? Answer: _____

P4. How many equal halves make a whole? Answer: _____

P5. How many equal quarters make a whole? Answer: _____

CHALLENGE PRACTICE

C1. Which is larger for the same whole: one-half or one-quarter? Answer: _____

C2. Two equal quarters together make what part? Answer: _____

C3. What makes a part a true half? Answer: _____

C4. What makes a part a true quarter? Answer: _____

C5. A whole has three quarters. Which part is missing? Answer: _____

C6. A whole has one half. Which part is missing? Answer: _____

C7. How many quarter-pieces cover the whole circle? Answer: _____

C8. One-quarter is one part out of how many equal pieces? Answer: _____

C9. One-half is one part out of how many equal pieces? Answer: _____

C10. Four equal quarters joined make what? Answer: _____

REAL-LIFE INVESTIGATION

1. Fold a paper circle into two equal parts, then four.

2. Sort drawings into equal and unequal partitions.

AT HOME

Share a paper rectangle equally between four people.

THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 14 | Count halves and quarters

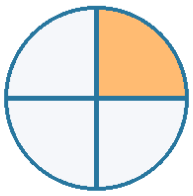
YOUR MISSION

Count equal pieces using the words one-half, two-halves and one-quarter, two-quarters and so on.

MATHS QUEST • BASIC 2 • LESSON 14

Count halves and quarters

Count the shaded pieces in a whole cut into 4 equal parts. What do we call 1 shaded piece? (Set 1)



1 of 4 equal parts shaded

SOLUTION: one-quarter

LEARN IT - FOUR IMPORTANT IDEAS

1. Each half is one of two equal pieces. Two halves join to make one whole.

2. Each quarter is one of four equal pieces. Count them: one-quarter, two-quarters, three-quarters, four-quarters.

3. Two quarters cover the same amount as one-half of the same whole.

4. Use equal-sized paper pieces. Put the pieces together to check how much of the whole they cover.

WORKED EXAMPLES

Try this	How it works
How many quarters make a whole?	Four
How many halves make a whole?	Two
Two quarters form	One-half

TRY IT YOURSELF

P1. Count the shaded pieces in a whole cut into 4 equal parts. What do we call 1 shaded piece? (Set 1) Answer: _____

P2. Count the shaded pieces in a whole cut into 4 equal parts. What do we call 2 shaded pieces? (Set 2) Answer: _____

P3. Count the shaded pieces in a whole cut into 4 equal parts. What do we call 3 shaded pieces? (Set 3) Answer: _____

P4. Count the shaded pieces in a whole cut into 4 equal parts. What do we call 4 shaded pieces? (Set 4) Answer: _____

P5. Count the shaded pieces in a whole cut into 2 equal parts. What do we call 1 shaded piece? (Set 5) Answer: _____

CHALLENGE PRACTICE

C1. Count the shaded pieces in a whole cut into 2 equal parts. What do we call 2 shaded pieces? (Set 6) Answer: _____

C2. Count the shaded pieces in a whole cut into 4 equal parts. What do we call 1 shaded piece? (Set 7) Answer: _____

C3. Count the shaded pieces in a whole cut into 4 equal parts. What do we call 2 shaded pieces? (Set 8) Answer: _____

C4. Count the shaded pieces in a whole cut into 4 equal parts. What do we call 3 shaded pieces? (Set 9) Answer: _____

C5. Count the shaded pieces in a whole cut into 4 equal parts. What do we call 4 shaded pieces? (Set 10) Answer: _____

C6. Count the shaded pieces in a whole cut into 2 equal parts. What do we call 1 shaded piece? (Set 11) Answer: _____

C7. Count the shaded pieces in a whole cut into 2 equal parts. What do we call 2 shaded pieces? (Set 12) Answer: _____

C8. Count the shaded pieces in a whole cut into 4 equal parts. What do we call 2 shaded pieces? (Set 13) Answer: _____

C9. Count the shaded pieces in a whole cut into 4 equal parts. What do we call 3 shaded pieces? (Set 14) Answer: _____

C10. Count the shaded pieces in a whole cut into 4 equal parts. What do we call 4 shaded pieces? (Set 15) Answer: _____

REAL-LIFE INVESTIGATION

1. Make four equal paper pieces and count them aloud.

2. Build a whole from two matching half-pieces.

AT HOME

Draw a whole divided into four equal parts.

THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 15 | Build wholes from equal parts

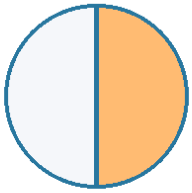
YOUR MISSION

Determine how many half-pieces or quarter-pieces are needed to make a whole.

MATHS QUEST • BASIC 2 • LESSON 15

Build wholes from equal parts

A whole is divided into 2 equal halves. We already have 1 of the parts. How many more equal parts make one whole? (Puzzle 1)



1 of 2 equal parts shaded

SOLUTION: 1

LEARN IT - FOUR IMPORTANT IDEAS

1. A whole can be made from two equal halves. If only one half is present, one more is needed.

2. A whole can be made from four equal quarters. If three quarters are present, one more is needed.

3. Quarter-pieces must be taken from the same-sized whole. Four tiny pieces from different shapes may not form the intended whole.

4. You can test a fraction statement with folded paper, a circle diagram or equal-sized blocks.

WORKED EXAMPLES

Try this	How it works
Three quarters plus how many?	One-quarter
One-half plus one-half	One whole
One quarter and three quarters	One whole

TRY IT YOURSELF

P1. A whole is divided into 2 equal halves. We already have 1 of the parts. How many more equal parts make one whole? (Puzzle 1) Answer: _____

P2. A whole is divided into 4 equal quarters. We already have 1 of the parts. How many more equal parts make one whole? (Puzzle 2) Answer: _____

P3. A whole is divided into 4 equal quarters. We already have 2 of the parts. How many more equal parts make one whole? (Puzzle 3) Answer: _____

P4. A whole is divided into 2 equal halves. We already have 1 of the parts. How many more equal parts make one whole? (Puzzle 4) Answer: _____

P5. A whole is divided into 4 equal quarters. We already have 1 of the parts. How many more equal parts make one whole? (Puzzle 5) Answer: _____

CHALLENGE PRACTICE

C1. A whole is divided into 4 equal quarters. We already have 1 of the parts. How many more equal parts make one whole? (Puzzle 6) Answer: _____

C2. A whole is divided into 2 equal halves. We already have 1 of the parts. How many more equal parts make one whole? (Puzzle 7) Answer: _____

C3. A whole is divided into 4 equal quarters. We already have 3 of the parts. How many more equal parts make one whole? (Puzzle 8) Answer: _____

C4. A whole is divided into 4 equal quarters. We already have 1 of the parts. How many more equal parts make one whole? (Puzzle 9) Answer: _____

C5. A whole is divided into 2 equal halves. We already have 1 of the parts. How many more equal parts make one whole? (Puzzle 10) Answer: _____

C6. A whole is divided into 4 equal quarters. We already have 2 of the parts. How many more equal parts make one whole? (Puzzle 11) Answer: _____

C7. A whole is divided into 4 equal quarters. We already have 3 of the parts. How many more equal parts make one whole? (Puzzle 12) Answer: _____

C8. A whole is divided into 2 equal halves. We already have 1 of the parts. How many more equal parts make one whole? (Puzzle 13) Answer: _____

C9. A whole is divided into 4 equal quarters. We already have 1 of the parts. How many more equal parts make one whole? (Puzzle 14) Answer: _____

C10. A whole is divided into 4 equal quarters. We already have 2 of the parts. How many more equal parts make one whole? (Puzzle 15) Answer: _____

REAL-LIFE INVESTIGATION

1. Rebuild a paper circle from four equal pieces.

2. Compare two-half and four-quarter representations.

AT HOME

Explain to a family member why equal pieces matter.

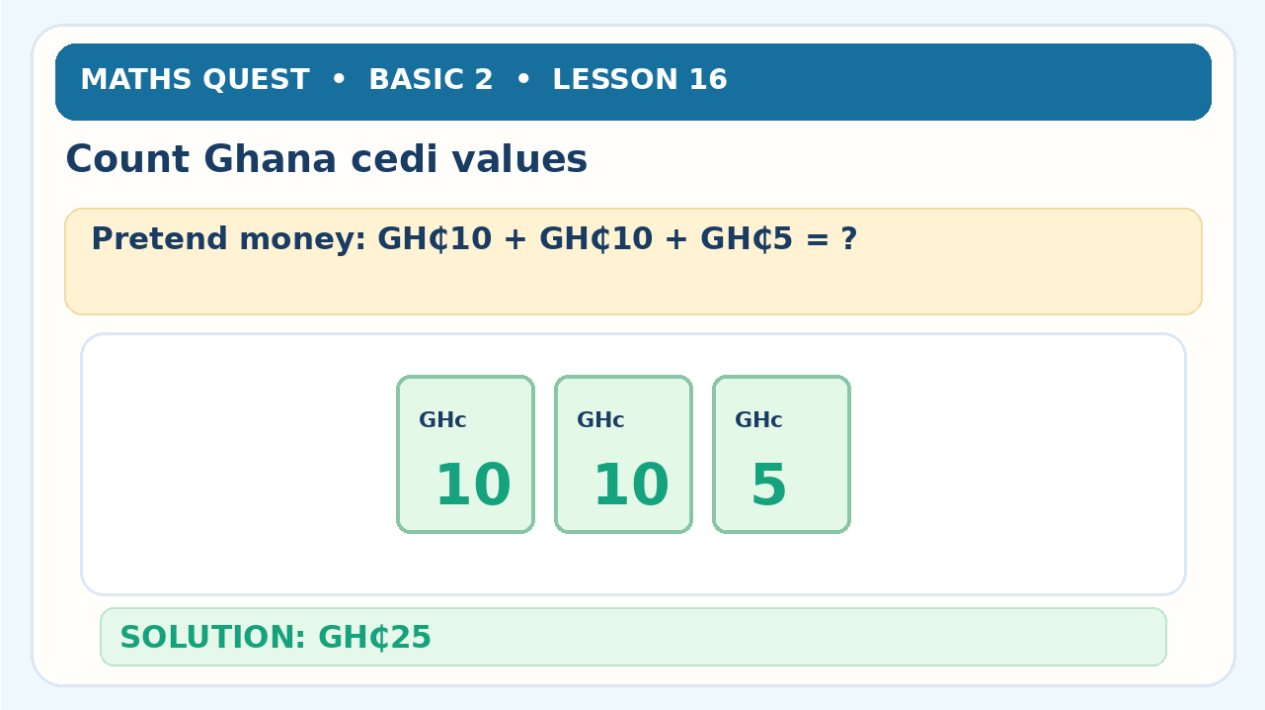
THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 16 | Count Ghana cedi values

YOUR MISSION

Recognise curriculum-specified cedi denominations and combine money values up to fifty Ghana cedis.



LEARN IT - FOUR IMPORTANT IDEAS

1. Money has different values. A five-cedi value represents more than a two-cedi value.

2. To count a collection, group equal denominations first and add the groups.

3. For example, two ten-cedi values and one five-cedi value make twenty-five cedis.

4. For practice, use printed pretend-money cards. Currency designs and circulation can change, so check actual notes with a trusted adult.

WORKED EXAMPLES

Try this	How it works
GH¢10 + GH¢10 + GH¢5	GH¢25
GH¢20 + GH¢5	GH¢25
Five GH¢2 values	GH¢10

TRY IT YOURSELF

P1. Pretend money: GH₵10 + GH₵10 + GH₵5 = ? Answer: _____

P2. Pretend money: GH₵20 + GH₵5 = ? Answer: _____

P3. Pretend money: GH₵2 + GH₵2 + GH₵2 = ? Answer: _____

P4. Pretend money: GH₵5 + GH₵5 + GH₵10 = ? Answer: _____

P5. Pretend money: GH₵10 + GH₵5 + GH₵2 = ? Answer: _____

CHALLENGE PRACTICE

C1. Pretend money: GH₵20 + GH₵20 + GH₵5 = ? Answer: _____

C2. Pretend money: GH₵5 + GH₵10 + GH₵10 = ? Answer: _____

C3. Pretend money: GH₵2 + GH₵2 + GH₵5 = ? Answer: _____

C4. Pretend money: GH₵10 + GH₵10 + GH₵10 = ? Answer: _____

C5. Pretend money: GH₵20 + GH₵10 + GH₵5 = ? Answer: _____

C6. Pretend money: GH₵5 + GH₵5 + GH₵5 = ? Answer: _____

C7. Pretend money: GH₵10 + GH₵2 + GH₵2 = ? Answer: _____

C8. Pretend money: GH₵20 + GH₵20 + GH₵10 = ? Answer: _____

C9. Pretend money: GH₵10 + GH₵5 + GH₵5 = ? Answer: _____

C10. Pretend money: GH₵2 + GH₵5 + GH₵10 = ? Answer: _____

REAL-LIFE INVESTIGATION

1. Make pretend money cards and buy classroom objects.

2. Combine three values in two different ways.

AT HOME

Plan a pretend purchase costing GH₵30.

THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 17 | Growing and shrinking patterns

YOUR MISSION

Recognise number patterns that increase or decrease by two, five and ten.



LEARN IT - FOUR IMPORTANT IDEAS

1. A number pattern follows a rule. Look at how each term changes.

2. In 4, 6, 8, 10 the rule is add two. In 30, 25, 20, 15 the rule is subtract five.

3. To find a missing term, identify the rule and keep applying it.

4. If a term breaks the rule, explain why it does not belong and replace it with the correct term.

WORKED EXAMPLES

Try this	How it works
4, 6, 8, __	10
30, 25, 20, __	15
50, 60, 70, __	80

TRY IT YOURSELF

P1. What is next? 10, 12, 14, 16, __ Answer: _____

P2. What is next? 10, 15, 20, 25, __ Answer: _____

P3. What is next? 10, 20, 30, 40, __ Answer: _____

P4. What is next? 22, 20, 18, 16, __ Answer: _____

P5. What is next? 22, 17, 12, 7, __ Answer: _____

CHALLENGE PRACTICE

C1. What is next? 30, 32, 34, 36, __ Answer: _____

C2. What is next? 30, 35, 40, 45, __ Answer: _____

C3. What is next? 30, 40, 50, 60, __ Answer: _____

C4. What is next? 42, 40, 38, 36, __ Answer: _____

C5. What is next? 42, 37, 32, 27, __ Answer: _____

C6. What is next? 50, 52, 54, 56, __ Answer: _____

C7. What is next? 50, 55, 60, 65, __ Answer: _____

C8. What is next? 50, 60, 70, 80, __ Answer: _____

C9. What is next? 62, 60, 58, 56, __ Answer: _____

C10. What is next? 62, 57, 52, 47, __ Answer: _____

REAL-LIFE INVESTIGATION

1. Use number cards to build three patterns.

2. Find the wrong term in a pattern and repair it.

AT HOME

Make an increasing pattern by fives.

THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 18 | Make and describe pattern rules

YOUR MISSION

Create, extend and explain patterns involving repeated addition or subtraction and equal arrays.

MATHS QUEST • BASIC 2 • LESSON 18

Make and describe pattern rules

An array has 2 rows of 3 counters. How many counters are in the array?



SOLUTION: 6

LEARN IT - FOUR IMPORTANT IDEAS

1. An array places objects in rows and columns. Equal rows can show a repeated grouping rule.

2. Patterns can add the same amount over and over: 2, 4, 6, 8 adds two each time.

3. Some patterns repeat a short group of things or numbers. Describe which part repeats.

4. You can create a pattern from a rule, or look at a pattern and work out its rule.

WORKED EXAMPLES

Try this	How it works
2, 4, 6, 8, __	10
20, 18, 16, __	14
Three rows of four objects	12 objects

TRY IT YOURSELF

P1. An array has 2 rows of 3 counters. How many counters are in the array? Answer: _____

P2. Follow the rule add 4: 9, 13, 17, __. What is missing? Answer: _____

P3. Follow the rule subtract 4: 66, 62, 58, __. What is missing? Answer: _____

P4. An array has 5 rows of 6 counters. How many counters are in the array? Answer: _____

P5. Follow the rule add 4: 21, 25, 29, __. What is missing? Answer: _____

CHALLENGE PRACTICE

C1. Follow the rule subtract 3: 75, 72, 69, __. What is missing? Answer: _____

C2. An array has 4 rows of 4 counters. How many counters are in the array? Answer: _____

C3. Follow the rule add 4: 33, 37, 41, __. What is missing? Answer: _____

C4. Follow the rule subtract 2: 84, 82, 80, __. What is missing? Answer: _____

C5. An array has 3 rows of 7 counters. How many counters are in the array? Answer: _____

C6. Follow the rule add 4: 45, 49, 53, __. What is missing? Answer: _____

C7. Follow the rule subtract 5: 93, 88, 83, __. What is missing? Answer: _____

C8. An array has 2 rows of 5 counters. How many counters are in the array? Answer: _____

C9. Follow the rule add 4: 57, 61, 65, __. What is missing? Answer: _____

C10. Follow the rule subtract 4: 102, 98, 94, __. What is missing? Answer: _____

REAL-LIFE INVESTIGATION

1. Make a three-by-four array with buttons.

2. Describe your repeating pattern to a partner.

AT HOME

Continue two number sequences from your workbook.

THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 19 | Recognise three-dimensional objects

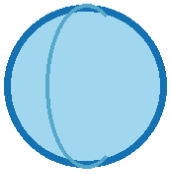
YOUR MISSION

Identify cubes, spheres, cylinders, cones and pyramids by their features.

MATHS QUEST • BASIC 2 • LESSON 19

Recognise three-dimensional objects

Which 3D object looks like a ball?



Sphere

SOLUTION: sphere

LEARN IT - FOUR IMPORTANT IDEAS

1. A three-dimensional object has length, width and height. You can hold a real 3D object.

2. A cube has six square faces. A sphere is round like a ball. A cylinder has two flat circular ends and a curved surface.

3. A cone has a circular base and a point. A pyramid has a base and triangular side faces that meet near the top.

4. Turn objects around to notice their faces, edges, corners and curved surfaces.

WORKED EXAMPLES

Try this	How it works
Object shaped like a ball	Sphere
Object shaped like a tin can	Cylinder
Object with six square faces	Cube

TRY IT YOURSELF

P1. Which 3D object looks like a ball? Answer: _____

P2. Which solid looks like a tin can? Answer: _____

P3. Which solid has six square faces? Answer: _____

P4. Which solid has a circular base and one point? Answer: _____

P5. Which solid has triangular sides rising from a base? Answer: _____

CHALLENGE PRACTICE

C1. What shape is a soccer ball? Answer: _____

C2. Which object has flat circular ends? Answer: _____

C3. What solid can be made from six equal paper squares? Answer: _____

C4. Which solid rolls smoothly in every direction? Answer: _____

C5. Which solid can have a square base and four triangular faces? Answer: _____

C6. Which solid looks like an ice-cream cone? Answer: _____

C7. What solid has no flat faces? Answer: _____

C8. Which solid has a curved surface and two circles? Answer: _____

C9. A dice-shaped block is usually a __. Answer: _____

C10. A party hat can be shaped like a __. Answer: _____

REAL-LIFE INVESTIGATION

1. Sort clean safe classroom objects by shape.

2. Build a cube from a paper net with adult help.

AT HOME

Find a cylinder and a sphere at home.

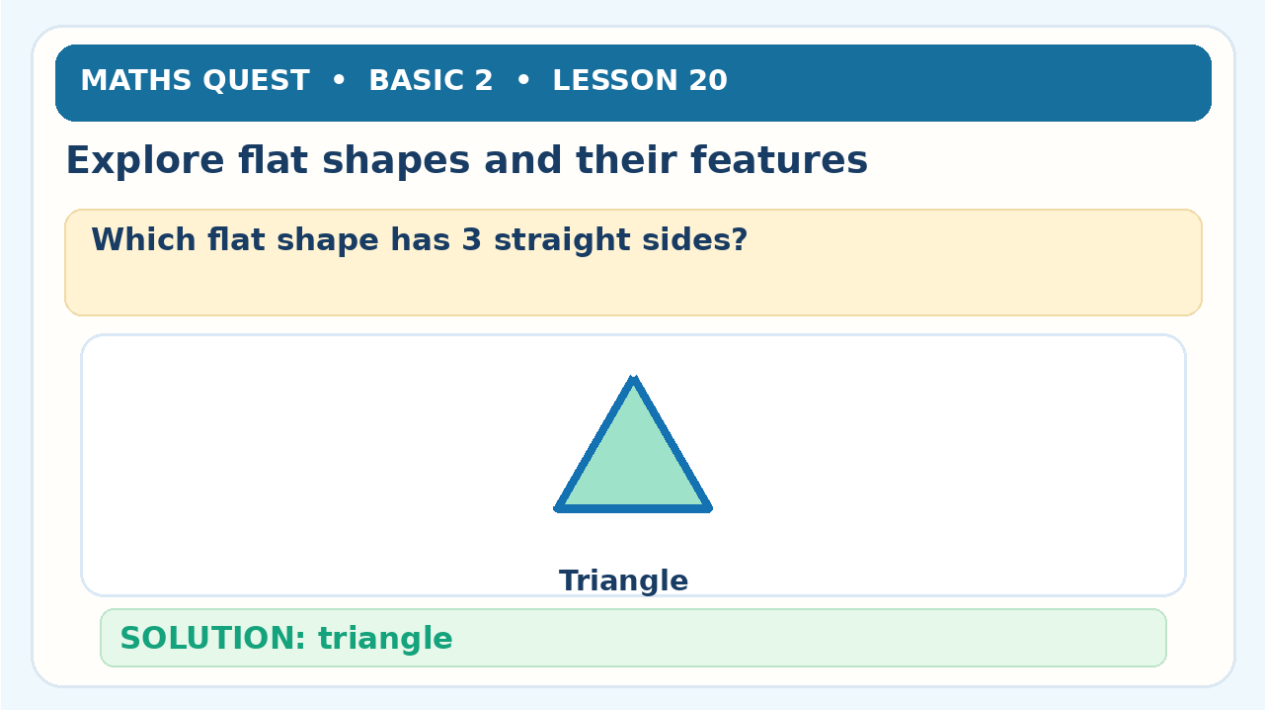
THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 20 | Explore flat shapes and their features

YOUR MISSION

Describe squares, rectangles, triangles, circles, pentagons and hexagons.



LEARN IT - FOUR IMPORTANT IDEAS

1. Two-dimensional shapes are flat. We can draw them on paper.

2. A triangle has three straight sides; a square has four equal straight sides; a rectangle has four straight sides.

3. A pentagon has five straight sides and a hexagon has six. A circle has no straight sides or corners.

4. A shape does not change its name when it is turned. Count sides and corners carefully.

WORKED EXAMPLES

Try this	How it works
Five-sided shape	Pentagon
Six-sided shape	Hexagon
Three-sided shape	Triangle

TRY IT YOURSELF

P1. Which flat shape has 3 straight sides? Answer: _____

P2. Which flat shape has 5 straight sides? Answer: _____

P3. Which flat shape has 6 straight sides? Answer: _____

P4. Which flat shape has no straight sides? Answer: _____

P5. Which shape has 4 equal sides and 4 corners? Answer: _____

CHALLENGE PRACTICE

C1. Which flat shape has 4 straight sides and 4 corners? Answer: _____

C2. A traffic warning sign can be a __. Answer: _____

C3. A round clock face is a __. Answer: _____

C4. A rectangular board has how many corners? Answer: _____

C5. A triangle has how many vertices? Answer: _____

C6. A hexagon has how many vertices? Answer: _____

C7. A pentagon has how many sides? Answer: _____

C8. A square has how many corners? Answer: _____

C9. A circle has how many corners? Answer: _____

C10. A shape turned sideways keeps its __. Answer: _____

REAL-LIFE INVESTIGATION

1. Draw each shape with labels.

2. Go on a safe shape hunt around school.

AT HOME

Find three rectangular objects at home.

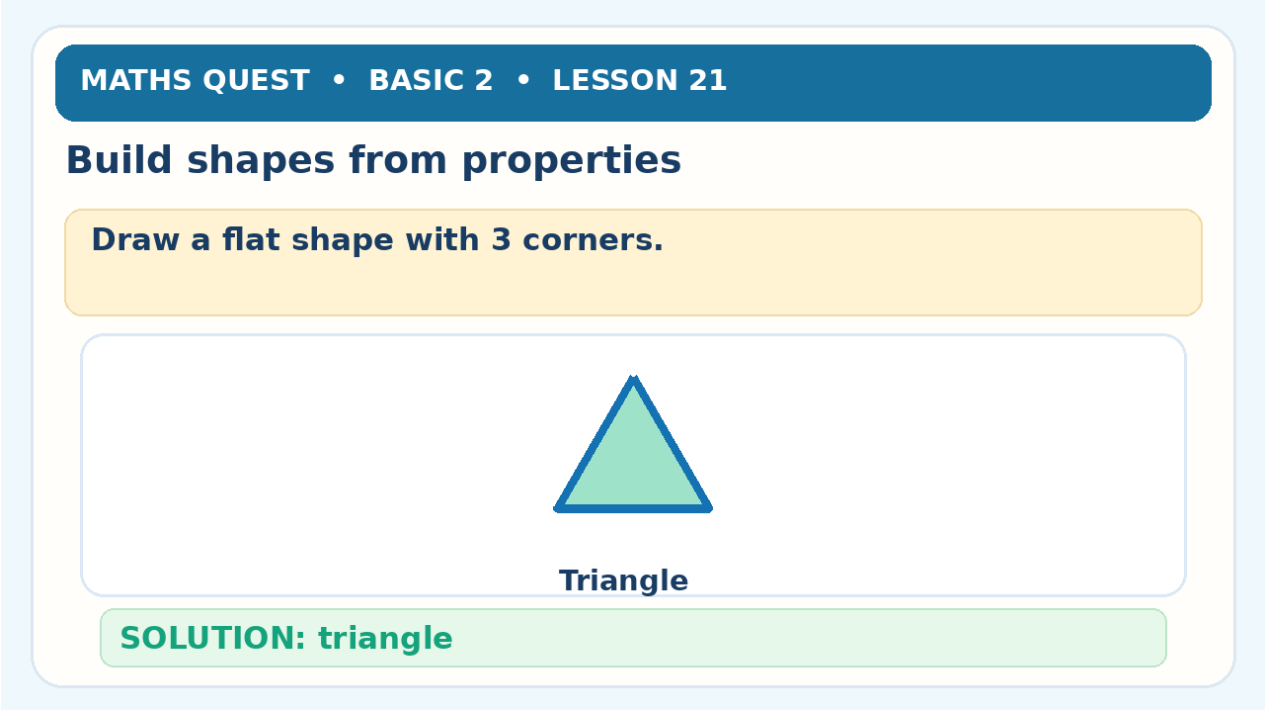
THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 21 | Build shapes from properties

YOUR MISSION

Draw flat shapes from clues about their sides and corners and find them on solids.



LEARN IT - FOUR IMPORTANT IDEAS

1. A side is a straight line segment around a polygon. A vertex is a corner where two sides meet.

2. If a clue says five straight sides, a pentagon is a good answer.

3. A cube has flat square faces. The top of a cylindrical tin can looks like a circle.

4. Use sticks, drawings and paper cut-outs to build and name shapes.

WORKED EXAMPLES

Try this	How it works
Shape with 5 corners	Pentagon
Flat face of a cube	Square
Circular end of a can	Circle

TRY IT YOURSELF

P1. Draw a flat shape with 3 corners. Answer: _____

P2. Draw a flat shape with 5 sides. Answer: _____

P3. Draw a flat shape with 6 sides. Answer: _____

P4. What is the flat face of a cube? Answer: _____

P5. What is the flat end of a cylindrical can? Answer: _____

CHALLENGE PRACTICE

C1. What is a 4-sided shape with equal sides? Answer: _____

C2. How many corners does a triangle have? Answer: _____

C3. How many corners does a hexagon have? Answer: _____

C4. How many sides does a pentagon have? Answer: _____

C5. What flat shape can you trace from the bottom of a cup? Answer: _____

C6. A paper shape with 3 straight edges is a __. Answer: _____

C7. A flat shape with no vertices is a __. Answer: _____

C8. Which shape uses 4 equal sticks? Answer: _____

C9. A cube face is flat and usually a __. Answer: _____

C10. How many vertices does a square have? Answer: _____

REAL-LIFE INVESTIGATION

1. Use five sticks to make a pentagon.

2. Trace the end of a cylindrical container.

AT HOME

Draw a shape with six straight sides.

THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 22 | Same length in different directions

YOUR MISSION

Demonstrate that turning or moving an object does not change its length.

MATHS QUEST • BASIC 2 • LESSON 22

Same length in different directions

A stick is 6 centimetres long. It is turned by 15 degrees. How long is it now?



Same length, different directions

SOLUTION: 6

LEARN IT - FOUR IMPORTANT IDEAS

1. A pencil stays the same length when you turn it from horizontal to vertical.

2. Your eyes can be tricked by an object’s position. Measure or place objects side by side to compare fairly.

3. A stick angled across a page does not become longer or shorter simply because it is tilted.

4. To compare lengths fairly, line up one end of each object at the same starting point.

WORKED EXAMPLES

Try this	How it works
Turn a 12 cm stick	Still 12 cm
Which of two equal sticks is longer?	Neither
Best way to compare sticks	Line up one end

TRY IT YOURSELF

P1. A stick is 6 centimetres long. It is turned by 15 degrees. How long is it now? Answer: _____

P2. Two 7-centimetre sticks point in different directions. How much longer is one than the other? Answer: _____

P3. You rotate a 8-centimetre paper strip so it points upwards. What length will a ruler show? Answer: _____

P4. A stick is 9 centimetres long. It is turned by 60 degrees. How long is it now? Answer: _____

P5. Two 10-centimetre sticks point in different directions. How much longer is one than the other? Answer: _____

CHALLENGE PRACTICE

C1. You rotate a 11-centimetre paper strip so it points upwards. What length will a ruler show? Answer: _____

C2. A stick is 12 centimetres long. It is turned by 105 degrees. How long is it now? Answer: _____

C3. Two 13-centimetre sticks point in different directions. How much longer is one than the other? Answer: _____

C4. You rotate a 14-centimetre paper strip so it points upwards. What length will a ruler show? Answer: _____

C5. A stick is 15 centimetres long. It is turned by 150 degrees. How long is it now? Answer: _____

C6. Two 16-centimetre sticks point in different directions. How much longer is one than the other? Answer: _____

C7. You rotate a 17-centimetre paper strip so it points upwards. What length will a ruler show? Answer: _____

C8. A stick is 18 centimetres long. It is turned by 15 degrees. How long is it now? Answer: _____

C9. Two 6-centimetre sticks point in different directions. How much longer is one than the other? Answer: _____

C10. You rotate a 7-centimetre paper strip so it points upwards. What length will a ruler show? Answer: _____

REAL-LIFE INVESTIGATION

1. Rotate a strip of paper and compare its length.

2. Put two equal straws in different directions.

AT HOME

Explain to a friend why turning does not change length.

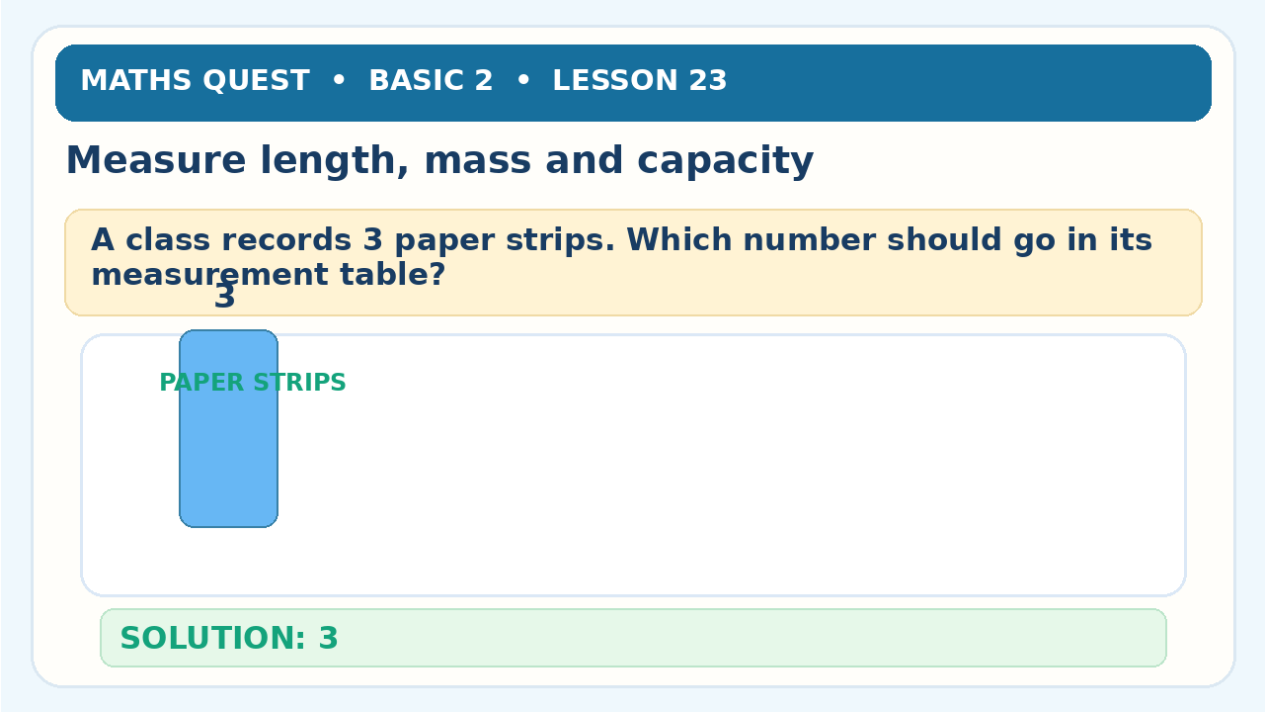
THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 23 | Measure length, mass and capacity

YOUR MISSION

Measure and compare everyday things using repeated non-standard units.



LEARN IT - FOUR IMPORTANT IDEAS

1. Measure length with identical paper clips or equal handspans, carefully placed end to end.

2. Mass tells us how heavy something is. A simple balance can compare two safe objects.

3. Capacity tells us how much a container can hold. Count equal cupfuls to compare larger containers.

4. Choose a sensible measuring unit. A tiny object is easier to measure with small units than with giant footsteps.

WORKED EXAMPLES

Try this	How it works
A shelf is 8 equal paper strips long	8 strips
A bucket holds 6 equal cupfuls	6 cupfuls
One side of a balance goes down	That side is heavier

TRY IT YOURSELF

P1. A class records 3 paper strips. Which number should go in its measurement table? Answer: _____

P2. A class records 4 handspans. Which number should go in its measurement table? Answer: _____

P3. A class records 5 equal cupfuls. Which number should go in its measurement table? Answer: _____

P4. A class records 6 equal blocks. Which number should go in its measurement table? Answer: _____

P5. A class records 7 paper clips. Which number should go in its measurement table? Answer: _____

CHALLENGE PRACTICE

C1. A class records 8 paper strips. Which number should go in its measurement table? Answer: _____

C2. A class records 9 handspans. Which number should go in its measurement table? Answer: _____

C3. A class records 10 equal cupfuls. Which number should go in its measurement table? Answer: _____

C4. A class records 3 equal blocks. Which number should go in its measurement table? Answer: _____

C5. A class records 4 paper clips. Which number should go in its measurement table? Answer: _____

C6. A class records 5 paper strips. Which number should go in its measurement table? Answer: _____

C7. A class records 6 handspans. Which number should go in its measurement table? Answer: _____

C8. A class records 7 equal cupfuls. Which number should go in its measurement table? Answer: _____

C9. A class records 8 equal blocks. Which number should go in its measurement table? Answer: _____

C10. A class records 9 paper clips. Which number should go in its measurement table? Answer: _____

REAL-LIFE INVESTIGATION

1. Make a paper-strip ruler with equal units.

2. Measure and compare two safe containers.

AT HOME

Record the lengths of a pencil and a book in paper clips.

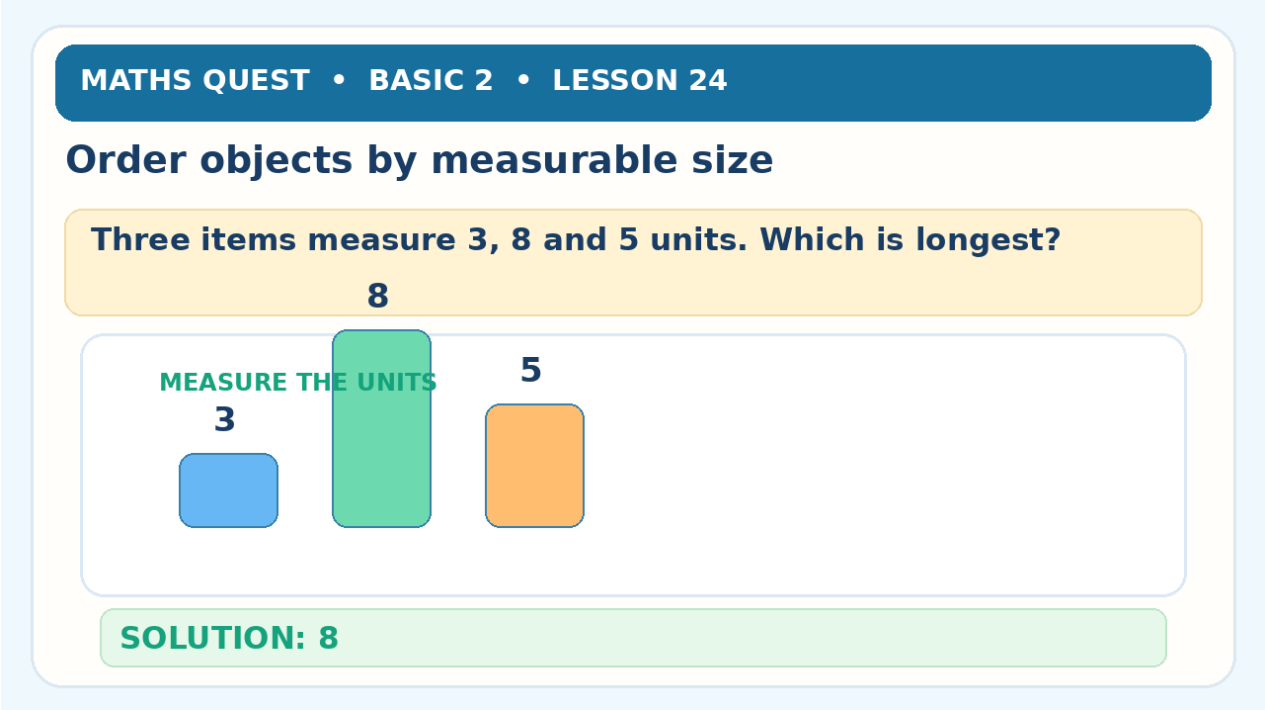
THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 24 | Order objects by measurable size

YOUR MISSION

Compare and arrange at least three items using size, length, mass and capacity words.



LEARN IT - FOUR IMPORTANT IDEAS

1. Compare objects using the same property each time. For length, use shortest, longer and longest.

2. For mass, use lighter, heavier and heaviest. For capacity, use less, more and most.

3. We can compare directly by lining objects up, or indirectly by using a measuring unit.

4. Record your measurements in a table before placing objects in order.

WORKED EXAMPLES

Try this	How it works
Lengths 3, 7, 5 strips: longest	7 strips
Masses 2, 4, 3 blocks: lightest	2 blocks
Cupfuls 4, 6, 5: greatest capacity	6 cupfuls

TRY IT YOURSELF

P1. Three items measure 3, 8 and 5 units. Which is longest? Answer: _____

P2. Three items measure 4, 9 and 6 units. Which is shortest? Answer: _____

P3. Three items measure 5, 10 and 7 units. Which is middle? Answer: _____

P4. Three items measure 6, 11 and 5 units. Which is longest? Answer: _____

P5. Three items measure 3, 12 and 6 units. Which is shortest? Answer: _____

CHALLENGE PRACTICE

C1. Three items measure 4, 8 and 7 units. Which is middle? Answer: _____

C2. Three items measure 5, 9 and 5 units. Which is longest? Answer: _____

C3. Three items measure 6, 10 and 6 units. Which is shortest? Answer: _____

C4. Three items measure 3, 11 and 7 units. Which is middle? Answer: _____

C5. Three items measure 4, 12 and 5 units. Which is longest? Answer: _____

C6. Three items measure 5, 8 and 6 units. Which is shortest? Answer: _____

C7. Three items measure 6, 9 and 7 units. Which is middle? Answer: _____

C8. Three items measure 3, 10 and 5 units. Which is longest? Answer: _____

C9. Three items measure 4, 11 and 6 units. Which is shortest? Answer: _____

C10. Three items measure 5, 12 and 7 units. Which is middle? Answer: _____

REAL-LIFE INVESTIGATION

1. Measure three pencils and order them.

2. Compare three cups with equal scoops of rice.

AT HOME

Make a table comparing three safe household objects.

THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 25 | Why standard units matter

YOUR MISSION

Explain why people need shared standard units to compare measurements accurately.

MATHS QUEST • BASIC 2 • LESSON 25

Why standard units matter

Which tool measures the length of a pencil fairly?



SOLUTION: a ruler

LEARN IT - FOUR IMPORTANT IDEAS

1. Handspans differ in size between children and adults. The same desk may be measured as five big handspans or seven smaller ones.

2. A standard unit has the same agreed size for everyone. A ruler uses centimetres to measure short lengths fairly.

3. Measure from the zero mark and read carefully. A unit should not change halfway through a measurement.

4. Use informal units to explore, then a standard unit when the measurements need to be compared precisely.

WORKED EXAMPLES

Try this	How it works
Why may handspan counts differ?	Hands differ in size
Good tool for pencil length	Ruler
Where do we begin on a ruler?	At zero

TRY IT YOURSELF

P1. Which tool measures the length of a pencil fairly? Answer: _____

P2. Where should a centimetre measurement begin? Answer: _____

P3. What does a standard unit have for everyone? Answer: _____

P4. Why may two children get different handspan counts? Answer: _____

P5. Which unit is suitable for a short book length? Answer: _____

CHALLENGE PRACTICE

C1. Which unit is an informal personal measure? Answer: _____

C2. What should stay the same in fair measurement? Answer: _____

C3. Which tool has centimetre marks? Answer: _____

C4. Which is better for a precise line length? Answer: _____

C5. Where on a ruler should you align an object's start? Answer: _____

C6. A handspan is different for everyone because of __. Answer: _____

C7. A ruler helps people compare lengths using __. Answer: _____

C8. A standard unit is useful because it has the __. Answer: _____

C9. What makes a class measurement repeatable? Answer: _____

C10. What should you use to measure a 5-centimetre strip? Answer: _____

REAL-LIFE INVESTIGATION

1. Measure the same book with two people's handspans.

2. Use a ruler to measure a paper strip.

AT HOME

Explain why schools use centimetres.

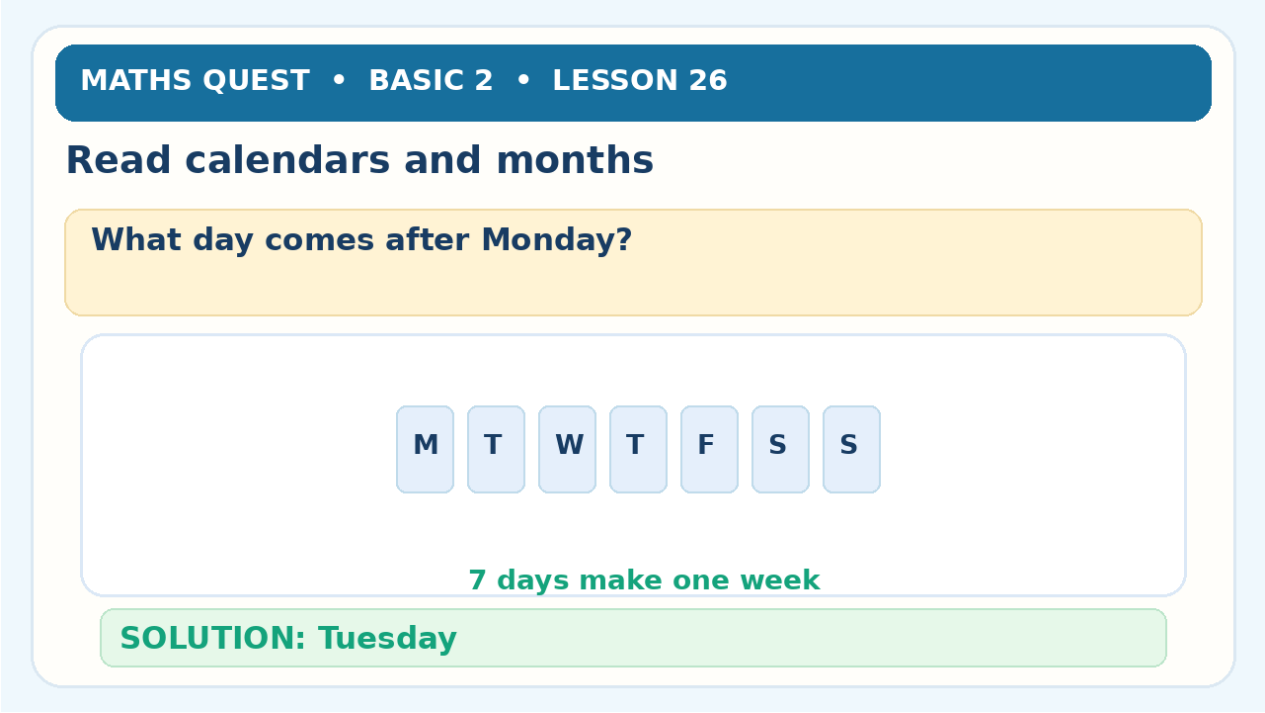
THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 26 | Read calendars and months

YOUR MISSION

Read days, weeks and months from a calendar and solve simple date questions.



LEARN IT - FOUR IMPORTANT IDEAS

1. A week has seven days: Monday, Tuesday, Wednesday, Thursday, Friday, Saturday and Sunday.

2. A year has twelve months. January comes first, February second and December last.

3. Most months have 30 or 31 days. February has 28 days in a common year and 29 in a leap year.

4. A calendar helps us find dates, days before and after, and how many days remain until an event.

WORKED EXAMPLES

Try this	How it works
Day after Tuesday	Wednesday
Month after April	May
Months in a year	12

TRY IT YOURSELF

P1. What day comes after Monday? Answer: _____

P2. What day comes after Tuesday? Answer: _____

P3. What day comes after Wednesday? Answer: _____

P4. What day comes after Thursday? Answer: _____

P5. What day comes after Friday? Answer: _____

CHALLENGE PRACTICE

C1. What day comes after Saturday? Answer: _____

C2. What day comes after Sunday? Answer: _____

C3. Which month comes after January? Answer: _____

C4. Which month comes after February? Answer: _____

C5. Which month comes after March? Answer: _____

C6. Which month comes after April? Answer: _____

C7. Which month comes after May? Answer: _____

C8. How many days are in a week? Answer: _____

C9. How many months are in a year? Answer: _____

C10. How many days does February have in a common year? Answer: _____

REAL-LIFE INVESTIGATION

1. Write all seven days in order.

2. Mark a special event on a calendar.

AT HOME

Find what day follows next Monday.

THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 27 | Hours and everyday events

YOUR MISSION

Describe duration using everyday timers and read whole hours on a clock.

MATHS QUEST • BASIC 2 • LESSON 27

Hours and everyday events

The minute hand points at 12. The hour hand points to 1. What time is it?



SOLUTION: 1 o’clock

LEARN IT - FOUR IMPORTANT IDEAS

1. A clock helps us tell time. When the long minute hand points to 12, we are at an exact hour.

2. The short hand points near the hour number. If the short hand points to 3 and the long hand to 12, it is three o’clock.

3. Different activities take different lengths of time. A song may take a few minutes while a school morning lasts several hours.

4. Estimate how long an event takes and compare with a clock or other safe timer.

WORKED EXAMPLES

Try this	How it works
Short hand 3, long hand 12	3 o’clock
An hour has how many minutes?	60
Day activity lasting longer	School morning

TRY IT YOURSELF

P1. The minute hand points at 12. The hour hand points to 1. What time is it? Answer: _____

P2. The minute hand points at 12. The hour hand points to 2. What time is it? Answer: _____

P3. The minute hand points at 12. The hour hand points to 3. What time is it? Answer: _____

P4. The minute hand points at 12. The hour hand points to 4. What time is it? Answer: _____

P5. The minute hand points at 12. The hour hand points to 5. What time is it? Answer: _____

CHALLENGE PRACTICE

C1. The minute hand points at 12. The hour hand points to 6. What time is it? Answer: _____

C2. The minute hand points at 12. The hour hand points to 7. What time is it? Answer: _____

C3. The minute hand points at 12. The hour hand points to 8. What time is it? Answer: _____

C4. The minute hand points at 12. The hour hand points to 9. What time is it? Answer: _____

C5. The minute hand points at 12. The hour hand points to 10. What time is it? Answer: _____

C6. The minute hand points at 12. The hour hand points to 11. What time is it? Answer: _____

C7. The minute hand points at 12. The hour hand points to 12. What time is it? Answer: _____

C8. How many minutes make one full hour? Answer: _____

C9. The long clock hand points to 12 at an exact hour. What is it called? Answer: _____

C10. Which hand points towards the number of the hour? Answer: _____

REAL-LIFE INVESTIGATION

1. Draw a paper clock with moving hands.

2. Compare a one-minute clap game and a short song.

AT HOME

Record the hour you start and finish a homework activity.

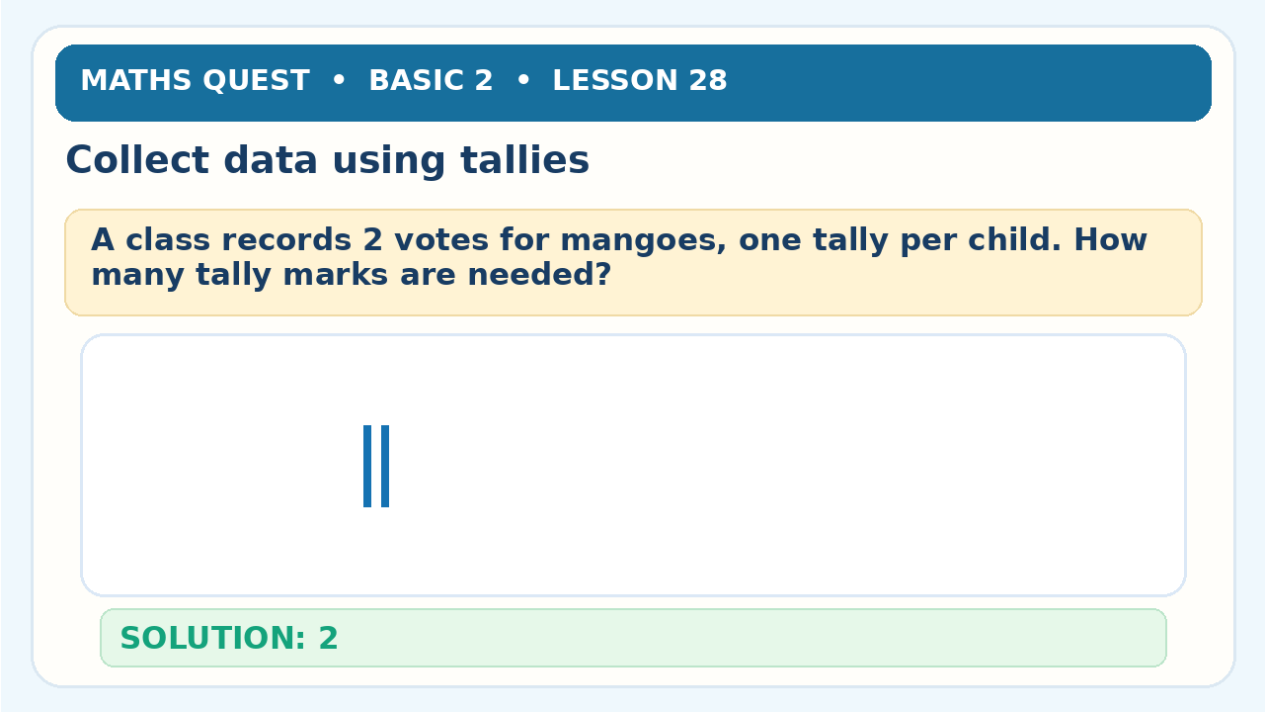
THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 28 | Collect data using tallies

YOUR MISSION

Ask questions, collect answers and organise them with tallies or checkmarks.



LEARN IT - FOUR IMPORTANT IDEAS

1. Data are pieces of information we gather. We might ask classmates about favourite fruits or sports.

2. Every answer must be recorded in its correct category. A tally mark counts one response.

3. For easier counting, we can group tally marks in fives. Four upright marks and a crossing fifth mark represent five.

4. Use a simple table to compare how many people chose each category. Check your total responses.

WORKED EXAMPLES

Try this	How it works
3 children choose mango	3 tally marks
A group of five tallies	5 responses
Which table records votes fairly?	One mark per vote

TRY IT YOURSELF

P1. A class records 2 votes for mangoes, one tally per child. How many tally marks are needed?
Answer: _____

P2. A class records 3 votes for mangoes, one tally per child. How many tally marks are needed?
Answer: _____

P3. A class records 4 votes for mangoes, one tally per child. How many tally marks are needed?
Answer: _____

P4. A class records 5 votes for mangoes, one tally per child. How many tally marks are needed?
Answer: _____

P5. A class records 6 votes for mangoes, one tally per child. How many tally marks are needed?
Answer: _____

CHALLENGE PRACTICE

C1. A class records 7 votes for mangoes, one tally per child. How many tally marks are needed?
Answer: _____

C2. A class records 8 votes for mangoes, one tally per child. How many tally marks are needed?
Answer: _____

C3. A class records 9 votes for mangoes, one tally per child. How many tally marks are needed?
Answer: _____

C4. A class records 10 votes for mangoes, one tally per child. How many tally marks are needed?
Answer: _____

C5. A class records 11 votes for mangoes, one tally per child. How many tally marks are needed?
Answer: _____

C6. A class records 12 votes for mangoes, one tally per child. How many tally marks are needed?
Answer: _____

C7. A class records 13 votes for mangoes, one tally per child. How many tally marks are needed?
Answer: _____

C8. A class records 14 votes for mangoes, one tally per child. How many tally marks are needed?
Answer: _____

C9. A class records 15 votes for mangoes, one tally per child. How many tally marks are needed?
Answer: _____

C10. A class records 16 votes for mangoes, one tally per child. How many tally marks are needed?
Answer: _____

REAL-LIFE INVESTIGATION

1. Survey five classmates about their favourite fruit.

2. Make a tally table with three categories.

AT HOME

Ask family members one question and record the responses.

THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 29 | Make and read picture graphs

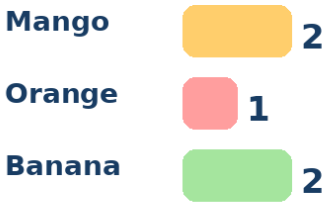
YOUR MISSION

Construct simple graphs or pictographs with up to three categories and answer questions about the data.

MATHS QUEST • BASIC 2 • LESSON 29

Make and read picture graphs

A graph shows mango 2, orange 1, banana 2. How many fruits altogether?



SOLUTION: 5

LEARN IT - FOUR IMPORTANT IDEAS

1. A pictograph uses pictures or symbols to show information clearly.

2. Read the key before counting. If one picture stands for one item, count the pictures. Some graphs use a picture to represent more than one item.

3. Put categories on one side and pictures in neat rows so the graph is easy to compare.

4. Read the graph to find the most, the least, how many altogether and the difference between groups.

WORKED EXAMPLES

Try this	How it works
Mango 4, orange 2, banana 3: most	Mango
4 mango + 2 orange	6 fruits
Difference between 4 and 2	2 fruits

TRY IT YOURSELF

P1. A graph shows mango 2, orange 1, banana 2. How many fruits altogether? Answer: _____

P2. A graph has 3 mangoes and 3 oranges. How many more is the larger group? Answer: _____

P3. One picture means one item. There are 3 banana pictures. How many bananas? Answer: _____

P4. A graph shows mango 5, orange 2, banana 6. How many fruits altogether? Answer: _____

P5. A graph has 6 mangoes and 4 oranges. How many more is the larger group? Answer: _____

CHALLENGE PRACTICE

C1. One picture means one item. There are 2 banana pictures. How many bananas? Answer: _____

C2. A graph shows mango 3, orange 3, banana 5. How many fruits altogether? Answer: _____

C3. A graph has 4 mangoes and 5 oranges. How many more is the larger group? Answer: _____

C4. One picture means one item. There are 6 banana pictures. How many bananas? Answer: _____

C5. A graph shows mango 6, orange 4, banana 4. How many fruits altogether? Answer: _____

C6. A graph has 2 mangoes and 1 oranges. How many more is the larger group? Answer: _____

C7. One picture means one item. There are 5 banana pictures. How many bananas? Answer: _____

C8. A graph shows mango 4, orange 5, banana 3. How many fruits altogether? Answer: _____

C9. A graph has 5 mangoes and 2 oranges. How many more is the larger group? Answer: _____

C10. One picture means one item. There are 4 banana pictures. How many bananas? Answer: _____

REAL-LIFE INVESTIGATION

1. Create a three-category pictograph from class votes.

2. Explain your graph to a friend.

AT HOME

Make a picture graph of favourite family foods.

THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Answer key for learner practice

Quest 01: Count, skip-count and estimate to 1,000

Play 1: 18 • Play 2: 50 • Play 3: 150 • Play 4: 50 • Play 5: 90

Challenge 1: 240 • Challenge 2: 62 • Challenge 3: 110 • Challenge 4: 370 • Challenge 5: 88 • Challenge 6: 170 • Challenge 7: 430 • Challenge 8: 70 • Challenge 9: 240 • Challenge 10: 630

Quest 02: Explore neighbours on number charts

Play 1: 28 • Play 2: 42 • Play 3: 76 • Play 4: 42 • Play 5: 85

Challenge 1: 14 • Challenge 2: 47 • Challenge 3: 61 • Challenge 4: 95 • Challenge 5: 45 • Challenge 6: 73 • Challenge 7: 14 • Challenge 8: 90 • Challenge 9: 31 • Challenge 10: 68

Quest 03: Measure and count using everyday units

Play 1: 3 • Play 2: 4 • Play 3: 5 • Play 4: 6 • Play 5: 7

Challenge 1: 8 • Challenge 2: 9 • Challenge 3: 10 • Challenge 4: 11 • Challenge 5: 3 • Challenge 6: 4 • Challenge 7: 5 • Challenge 8: 6 • Challenge 9: 7 • Challenge 10: 8

Quest 04: Build numbers with hundreds, tens and ones

Play 1: 300 • Play 2: 70 • Play 3: 5 • Play 4: 600 • Play 5: 50

Challenge 1: 3 • Challenge 2: 300 • Challenge 3: 10 • Challenge 4: 2 • Challenge 5: 900 • Challenge 6: 30 • Challenge 7: 6 • Challenge 8: 200 • Challenge 9: 10 • Challenge 10: 4

Quest 05: Describe numbers in equivalent ways

Play 1: 150 • Play 2: 77 • Play 3: 245 • Play 4: 315 • Play 5: 98

Challenge 1: 180 • Challenge 2: 417 • Challenge 3: 58 • Challenge 4: 275 • Challenge 5: 150 • Challenge 6: 308 • Challenge 7: 66 • Challenge 8: 501 • Challenge 9: 181 • Challenge 10: 97

Quest 06: Compare and order numbers

Play 1: > • Play 2: > • Play 3: < • Play 4: = • Play 5: >

Challenge 1: < • Challenge 2: > • Challenge 3: > • Challenge 4: = • Challenge 5: < • Challenge 6: < • Challenge 7: > • Challenge 8: < • Challenge 9: > • Challenge 10: >

Quest 07: Discover addition and subtraction relationships

Play 1: 59 • Play 2: 18 • Play 3: 40 • Play 4: 44 • Play 5: 27

Challenge 1: 55 • Challenge 2: 47 • Challenge 3: 49 • Challenge 4: 14 • Challenge 5: 57 • Challenge 6: 62 • Challenge 7: 26 • Challenge 8: 55 • Challenge 9: 45 • Challenge 10: 81

Quest 08: Equal and not equal number sentences

Play 1: ≠ • Play 2: = • Play 3: = • Play 4: = • Play 5: =

Challenge 1: ≠ • Challenge 2: = • Challenge 3: = • Challenge 4: = • Challenge 5: ≠ • Challenge 6: = • Challenge 7: = • Challenge 8: = • Challenge 9: = • Challenge 10: ≠

Quest 09: Quick mental maths facts

Play 1: 13 • Play 2: 13 • Play 3: 13 • Play 4: 6 • Play 5: 15

Challenge 1: 13 • Challenge 2: 18 • Challenge 3: 7 • Challenge 4: 11 • Challenge 5: 12 • Challenge 6: 14 • Challenge 7: 5 • Challenge 8: 17 • Challenge 9: 16 • Challenge 10: 11

Quest 10: Add and subtract using tens and ones

Play 1: 45 • Play 2: 26 • Play 3: 63 • Play 4: 71 • Play 5: 27

Challenge 1: 75 • Challenge 2: 37 • Challenge 3: 73 • Challenge 4: 27 • Challenge 5: 73 • Challenge 6: 37 • Challenge 7: 75 • Challenge 8: 53 • Challenge 9: 71 • Challenge 10: 28

Quest 11: Friendly jumps and clever strategies

Play 1: 75 • Play 2: 58 • Play 3: 85 • Play 4: 63 • Play 5: 75

Challenge 1: 80 • Challenge 2: 59 • Challenge 3: 81 • Challenge 4: 82 • Challenge 5: 79 • Challenge 6: 61 • Challenge 7: 80 • Challenge 8: 80 • Challenge 9: 80 • Challenge 10: 77

Quest 12: Solve everyday word problems

Play 1: 28 • Play 2: 22 • Play 3: 31 • Play 4: 37 • Play 5: 28

Challenge 1: 40 • Challenge 2: 46 • Challenge 3: 34 • Challenge 4: 49 • Challenge 5: 46 • Challenge 6: 40 • Challenge 7: 49 • Challenge 8: 55 • Challenge 9: 46 • Challenge 10: 58

Quest 13: Explore halves and quarters

Play 1: one-half • Play 2: one-quarter • Play 3: No • Play 4: Two • Play 5: Four

Challenge 1: one-half • Challenge 2: one-half • Challenge 3: Two equal parts • Challenge 4: Four equal parts • Challenge 5: one-quarter • Challenge 6: one-half • Challenge 7: Four • Challenge 8: Four • Challenge 9: Two • Challenge 10: one whole

Quest 14: Count halves and quarters

Play 1: one-quarter • Play 2: two-quarters • Play 3: three-quarters • Play 4: four-quarters • Play 5: one-half

Challenge 1: two halves • Challenge 2: one-quarter • Challenge 3: two-quarters • Challenge 4: three-quarters • Challenge 5: four-quarters • Challenge 6: one-half • Challenge 7: two halves • Challenge 8: two-quarters • Challenge 9: three-quarters • Challenge 10: four-quarters

Quest 15: Build wholes from equal parts

Play 1: 1 • Play 2: 3 • Play 3: 2 • Play 4: 1 • Play 5: 3

Challenge 1: 3 • Challenge 2: 1 • Challenge 3: 1 • Challenge 4: 3 • Challenge 5: 1 • Challenge 6: 2 • Challenge 7: 1 • Challenge 8: 1 • Challenge 9: 3 • Challenge 10: 2

Quest 16: Count Ghana cedi values

Play 1: GH₵25 • Play 2: GH₵25 • Play 3: GH₵6 • Play 4: GH₵20 • Play 5: GH₵17

Challenge 1: GH₵45 • Challenge 2: GH₵25 • Challenge 3: GH₵9 • Challenge 4: GH₵30 • Challenge 5: GH₵35 • Challenge 6: GH₵15 • Challenge 7: GH₵14 • Challenge 8: GH₵50 • Challenge 9: GH₵20 • Challenge 10: GH₵17

Quest 17: Growing and shrinking patterns

Play 1: 18 • Play 2: 30 • Play 3: 50 • Play 4: 14 • Play 5: 2

Challenge 1: 38 • Challenge 2: 50 • Challenge 3: 70 • Challenge 4: 34 • Challenge 5: 22 • Challenge 6: 58 • Challenge 7: 70 • Challenge 8: 90 • Challenge 9: 54 • Challenge 10: 42

Quest 18: Make and describe pattern rules

Play 1: 6 • Play 2: 21 • Play 3: 54 • Play 4: 30 • Play 5: 33

Challenge 1: 66 • Challenge 2: 16 • Challenge 3: 45 • Challenge 4: 78 • Challenge 5: 21 • Challenge 6: 57 • Challenge 7: 78 • Challenge 8: 10 • Challenge 9: 69 • Challenge 10: 90

Quest 19: Recognise three-dimensional objects

Play 1: sphere • Play 2: cylinder • Play 3: cube • Play 4: cone • Play 5: pyramid

Challenge 1: sphere • Challenge 2: cylinder • Challenge 3: cube • Challenge 4: sphere • Challenge 5: pyramid • Challenge 6: cone • Challenge 7: sphere • Challenge 8: cylinder • Challenge 9: cube • Challenge 10: cone

Quest 20: Explore flat shapes and their features

Play 1: triangle • Play 2: pentagon • Play 3: hexagon • Play 4: circle • Play 5: square

Challenge 1: rectangle • Challenge 2: triangle • Challenge 3: circle • Challenge 4: 4 • Challenge 5: 3 • Challenge 6: 6 • Challenge 7: 5 • Challenge 8: 4 • Challenge 9: 0 • Challenge 10: name

Quest 21: Build shapes from properties

Play 1: triangle • Play 2: pentagon • Play 3: hexagon • Play 4: square • Play 5: circle

Challenge 1: square • Challenge 2: 3 • Challenge 3: 6 • Challenge 4: 5 • Challenge 5: circle • Challenge 6: triangle • Challenge 7: circle • Challenge 8: square • Challenge 9: square • Challenge 10: 4

Quest 22: Same length in different directions

Play 1: 6 • Play 2: 0 • Play 3: 8 • Play 4: 9 • Play 5: 0

Challenge 1: 11 • Challenge 2: 12 • Challenge 3: 0 • Challenge 4: 14 • Challenge 5: 15 • Challenge 6: 0 • Challenge 7: 17 • Challenge 8: 18 • Challenge 9: 0 • Challenge 10: 7

Quest 23: Measure length, mass and capacity

Play 1: 3 • Play 2: 4 • Play 3: 5 • Play 4: 6 • Play 5: 7

Challenge 1: 8 • Challenge 2: 9 • Challenge 3: 10 • Challenge 4: 3 • Challenge 5: 4 • Challenge 6: 5 • Challenge 7: 6 • Challenge 8: 7 • Challenge 9: 8 • Challenge 10: 9

Quest 24: Order objects by measurable size

Play 1: 8 • Play 2: 4 • Play 3: 7 • Play 4: 11 • Play 5: 3

Challenge 1: 7 • Challenge 2: 9 • Challenge 3: 6 • Challenge 4: 7 • Challenge 5: 12 • Challenge 6: 5 • Challenge 7: 7 • Challenge 8: 10 • Challenge 9: 4 • Challenge 10: 7

Quest 25: Why standard units matter

Play 1: a ruler • Play 2: the zero mark • Play 3: same agreed size • Play 4: different hand sizes • Play 5: a centimetre

Challenge 1: a handspan • Challenge 2: same agreed size • Challenge 3: a ruler • Challenge 4: a ruler • Challenge 5: the zero mark • Challenge 6: different hand sizes • Challenge 7: a centimetre • Challenge 8: same agreed size • Challenge 9: same agreed size • Challenge 10: a ruler

Quest 26: Read calendars and months

Play 1: Tuesday • Play 2: Wednesday • Play 3: Thursday • Play 4: Friday • Play 5: Saturday

Challenge 1: Sunday • Challenge 2: Monday • Challenge 3: February • Challenge 4: March • Challenge 5: April • Challenge 6: May • Challenge 7: June • Challenge 8: 7 • Challenge 9: 12 • Challenge 10: 28

Quest 27: Hours and everyday events

Play 1: 1 o’clock • Play 2: 2 o’clock • Play 3: 3 o’clock • Play 4: 4 o’clock • Play 5: 5 o’clock

Challenge 1: 6 o’clock • Challenge 2: 7 o’clock • Challenge 3: 8 o’clock • Challenge 4: 9 o’clock • Challenge 5: 10 o’clock • Challenge 6: 11 o’clock • Challenge 7: 12 o’clock • Challenge 8: 60 • Challenge 9: the minute hand • Challenge 10: the hour hand

Quest 28: Collect data using tallies

Play 1: 2 • Play 2: 3 • Play 3: 4 • Play 4: 5 • Play 5: 6

Challenge 1: 7 • Challenge 2: 8 • Challenge 3: 9 • Challenge 4: 10 • Challenge 5: 11 • Challenge 6: 12 • Challenge 7: 13 • Challenge 8: 14 • Challenge 9: 15 • Challenge 10: 16

Quest 29: Make and read picture graphs

Play 1: 5 • Play 2: 0 • Play 3: 3 • Play 4: 13 • Play 5: 2

Challenge 1: 2 • Challenge 2: 11 • Challenge 3: 1 • Challenge 4: 6 • Challenge 5: 14 • Challenge 6: 1 • Challenge 7: 5 • Challenge 8: 12 • Challenge 9: 3 • Challenge 10: 4